Engineering Challenges

Section Européenne – Sciences de l'Ingénieur

EC01 to EC30

Version 4.0

July 2026

Revision History

Version	Date	Main developments
V1	Initial release	First collection and initial chapter structure.
V2	Earlier revision	Layout, illustrations and graphical presentation improved.
V3	Earlier revision	Collection expanded and engineering orientation strengthened.
V4.0	**July 2026**	Complete editorial revision of EC01–EC30 for Terminale SI: B2/C1 technical English, measurable objectives, quantitative applications, stronger modelling and validation, and harmonised TikZ/PGFPlots figures.

V4.0 revision lots

Pull request	Chapters	Scope
#7	EC01–EC05	First editorial lot.
#8	EC06–EC10	Second editorial lot.
#9	EC11–EC15	Third editorial lot.
#10	EC16–EC20	Fourth editorial lot.
#11	EC21–EC25	Fifth editorial lot.
#12	EC26–EC30	Final editorial lot.

Contents

EC01 – Renewable Energy

Can solar power meet tomorrow's energy needs?

Learning Objectives

- Explain how a photovoltaic cell and a grid-connected PV system operate.
- Interpret capacity data and estimate annual production.
- Discuss intermittency, storage, grid integration and recycling.

Engineering News

Global renewable capacity grew rapidly in 2025, and solar power represented most of the new installations. The challenge is no longer limited to producing more panels: engineers must also connect variable generation to stronger, smarter and more flexible electricity grids.

Main Article

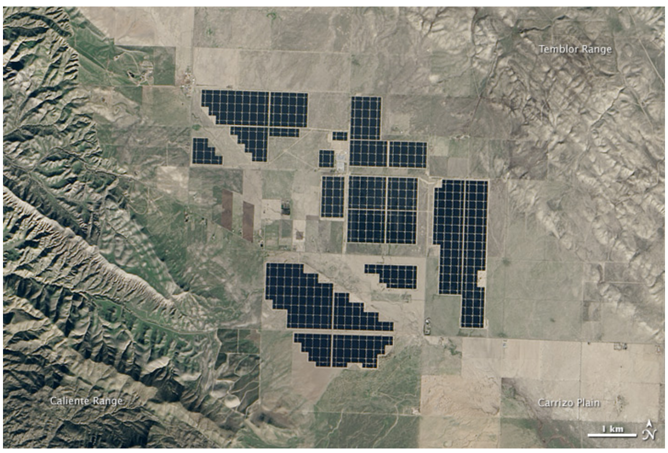
A photovoltaic (PV) cell converts light directly into electricity. Most commercial cells are made from silicon, a semiconductor. When photons transfer enough energy to the material, electrons can move. The internal electric field of the p–n junction separates positive and negative charges and creates a direct current.

A PV module connects many cells together. Modules are then combined into strings and arrays to reach the required voltage and power. However, a module does not operate at a fixed point. Its voltage-current characteristic changes with solar irradiance and temperature. A maximum power point tracker adjusts the operating point so that the array can deliver as much power as possible.

The generated electricity is direct current, whereas buildings and public grids use alternating current. An inverter therefore converts DC into AC, synchronises voltage and frequency with the grid, and provides protection and monitoring functions. A battery can store surplus electricity, but it also increases cost, material use and conversion losses.

Solar power is variable because output changes with clouds, seasons and time of day. Engineers integrate large amounts of PV through forecasting, flexible demand, interconnections, storage and controllable generation. The objective is not to make every building independent, but to balance the complete electricity system.

The environmental impact of a PV system must be assessed over its whole life cycle. Manufacturing requires energy and materials, while end-of-life modules contain glass, aluminium, copper, polymers and semiconductor materials. Designing panels that are efficient, durable, repairable and recyclable is therefore a major engineering objective.



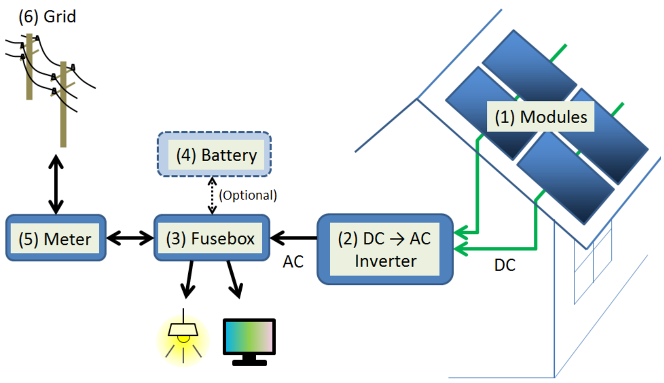
Topaz Solar Farm, California. NASA image, public domain.

Engineering Focus

Designing a photovoltaic installation
Engineers must optimise several parameters at the same time:

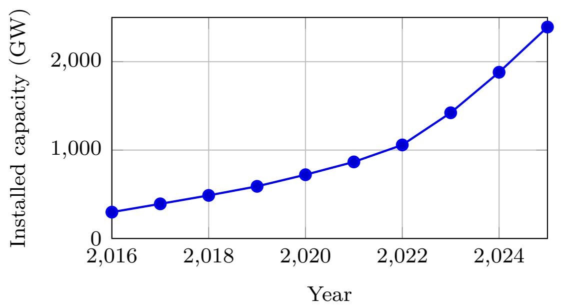
- panel orientation and tilt;
- inverter sizing and MPPT architecture;
- cable cross-sections and electrical losses;
- thermal behaviour and ventilation;
- protection and isolation devices;
- monitoring and maintenance strategy;
- expected lifetime and recyclability.

A good design must combine safety, reliability, high annual energy yield and acceptable cost.



Simplified residential PV system. S-kei, CC0.

Global solar PV capacity



Source: IRENA, Renewable Capacity Statistics 2026.

Engineering Insight

The Earth receives an enormous amount of solar energy, but useful production depends on available area, local weather, orientation, conversion efficiency and the ability of the grid to absorb variable power.

Key Figures

Commercial module efficiency	20–24%
Typical module lifetime	25–30 years
Typical inverter efficiency	97–99%
Annual module degradation	0.3–0.5%/year
Global PV capacity in 2025	≈ 2.4 TW

Useful Vocabulary

solar cell	cellule photovoltaïque
irradiance	irradiance
inverter	onduleur
grid	réseau électrique
storage	stockage
yield	production énergétique
lifetime	durée de vie
recycling	recyclage

Questions

1. Explain the role of the p–n junction.
2. Why does a PV installation need an inverter?
3. What is the purpose of MPPT control?
4. Use the graph to estimate the increase between 2020 and 2025.
5. Give three solutions for integrating variable solar power.
6. Why must environmental assessment include manufacturing and end-of-life?

Engineering Challenge

Your school consumes 410 MWh of electricity each year and has 850 m^2 of usable roof area. Prepare a preliminary PV study: estimate the peak power, estimate annual production, discuss self-consumption, decide whether a battery is justified and propose useful monitoring indicators.

Application Exercise

A 420 W PV module receives an average irradiance of 850 W/m^2 over an active area of 2.0 m^2. Calculate its instantaneous efficiency. Then estimate the electrical energy produced by 120 identical modules during 4.5 equivalent full-sun hours. State two reasons why the real daily yield may be lower.

Speaking Task

Working in pairs, prepare a five-minute presentation answering the question: “Can Europe rely mainly on solar electricity by 2050?” Support your arguments with technical evidence, environmental impacts and economic considerations.

EC02 – Hydrogen

Can hydrogen decarbonise heavy transport?

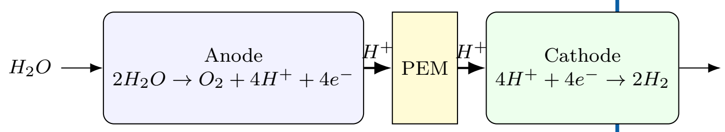
Learning Objectives

- Distinguish grey, blue and green hydrogen.
- Explain the operating principles of electrolysers and fuel cells.
- Compare hydrogen and battery-electric energy chains.

Engineering News

Low-emission hydrogen projects are being developed for fertiliser production, steelmaking, shipping and heavy road transport. Their success depends not only on electrolysers, but also on electricity supply, storage, distribution, safety and reliable demand.

PEM electrolyser



Main Article

Hydrogen is an **energy carrier**, not a primary source of energy. It must first be produced from another resource. Most hydrogen is still made from natural gas. When the carbon dioxide is released, it is called grey hydrogen. If part of the CO_2 is captured, it is often called blue hydrogen. Green hydrogen is produced by electrolysis using low-carbon renewable electricity.

In a proton-exchange-membrane (PEM) electrolyser, water enters the anode. An electric current separates water molecules into oxygen, protons and electrons. The membrane allows protons to pass through, whereas electrons travel through an external circuit. Hydrogen is formed at the cathode.

A PEM fuel cell performs the reverse overall reaction. Hydrogen reaches the anode and oxygen from the air reaches the cathode. The electrochemical reaction produces electricity, water and heat without combustion. Many cells are assembled into a stack to obtain the required voltage and power.

Hydrogen contains a large amount of energy per kilogram and can be refuelled quickly. However, it occupies a large volume and must be compressed, liquefied or converted into another carrier. Every conversion causes energy losses. Battery-electric vehicles are therefore generally more efficient, while hydrogen may remain attractive for long range, heavy payloads and short refuelling stops.

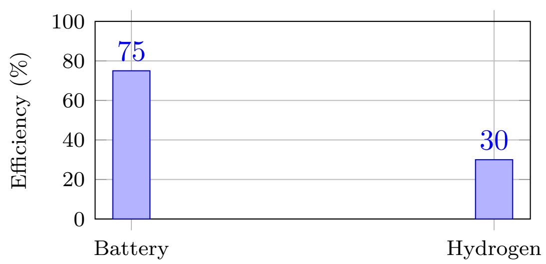
Engineering Focus

The complete hydrogen value chain

1. Generate low-carbon electricity.
2. Produce hydrogen by electrolysis.
3. Purify and compress the gas.
4. Store and distribute it safely.
5. Convert it into useful energy.

Engineers must optimise efficiency, safety, cost, materials and infrastructure at every stage.

Electricity-to-wheel efficiency



Illustrative values; actual performance depends on equipment and operating conditions.

Engineering Insight

The most efficient solution is not always the most suitable one. Range, payload, refuelling time, infrastructure and lifecycle emissions must also be considered.

Key Figures

Hydrogen LHV	≈ 120 MJ/kg
Electrolyser efficiency	60–75%
Fuel-cell electrical efficiency	40–60%
Vehicle storage pressure	350 or 700 bar

Useful Vocabulary

electrolyser	électrolyseur
fuel cell	pile à combustible
stack	empilement de cellules
compressed gas	gaz comprimé
refuelling	ravitaillement
leak detection	détection de fuite

Questions

1. Why is hydrogen described as an energy carrier?
2. Compare grey, blue and green hydrogen.
3. Explain the role of the PEM in an electrolyser.
4. Why is the battery chain usually more efficient?
5. In which transport applications can hydrogen remain attractive?

Engineering Challenge

A logistics company operates 120 long-haul trucks. Compare battery-electric and hydrogen fuel-cell solutions using efficiency, range, payload, refuelling time, infrastructure and lifecycle emissions. Present a justified recommendation.

Application Exercise

An electrolyser consumes 52 kWh of electricity to produce 1 kg of hydrogen. A fuel-cell system later converts the hydrogen into 18 kWh of useful electrical energy. Calculate the power-to-power efficiency. Compare it with an 88% battery storage system and explain why hydrogen may still be selected.

Speaking Task

Debate: "Hydrogen will replace batteries for heavy transport." Each team must use at least three technical arguments and respond to one opposing argument.

EC03 – Wind Energy

How do engineers harvest the wind?

Learning Objectives

- Explain how aerodynamic lift creates rotor torque.
- Identify the main subsystems of a modern wind turbine.
- Read and interpret a wind-turbine power curve.
- Compare onshore, fixed-bottom offshore and floating offshore solutions.

Engineering News

Modern offshore wind turbines can exceed 15 MW. Floating foundations open access to deeper-water sites, where winds are often stronger and steadier. They also create new engineering challenges involving moorings, dynamic cables, maintenance and grid connection.

Main Article

A wind turbine converts part of the kinetic energy of moving air into electrical energy. Its blades are shaped like aerofoils. As air flows around a blade, the pressure on one side becomes lower than on the other. This pressure difference creates a lift force. Because the blade is inclined and rotating, part of this lift acts tangentially and produces a torque on the rotor.

The power available in the wind is

$$P_{\text{wind}} = \frac{1}{2}\rho A v^3,$$

where ρ is air density, A the swept area and v wind speed. The cubic dependence on v is essential: doubling wind speed multiplies the available power by eight. This is why careful site measurement and long-term wind statistics are more important than a single instantaneous reading.

The rotor drives either a gearbox and a high-speed generator or a larger direct-drive generator. Power electronics adapt the variable electrical output to the frequency and voltage of the grid. Pitch control changes the angle of the blades to regulate power and limit loads. Yaw control turns the nacelle so that the rotor faces the wind.

A wind turbine does not produce its rated power all the time. Below the cut-in speed, it remains stopped. Between cut-in and rated speed, output rises rapidly. Above rated speed, control systems limit the power. At the cut-out speed, the turbine shuts down to avoid excessive mechanical loads.

Engineering Focus

Main subsystems

1. rotor, blades and hub;
2. main shaft and bearings;
3. gearbox or direct-drive generator;
4. converter and transformer;
5. pitch and yaw actuators;
6. tower, foundations and monitoring system.

A reliable design must combine high energy capture with acceptable fatigue loads, safe maintenance and a lifetime of several decades.

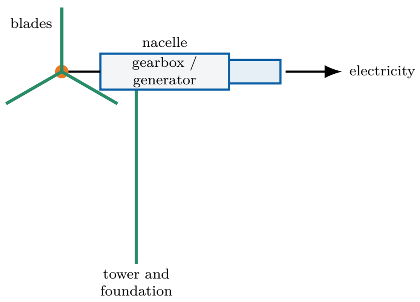
Did You Know?

No turbine can capture all the power in the wind. The theoretical upper limit is the **Betz limit**: 59.3% of the kinetic power crossing the rotor area.

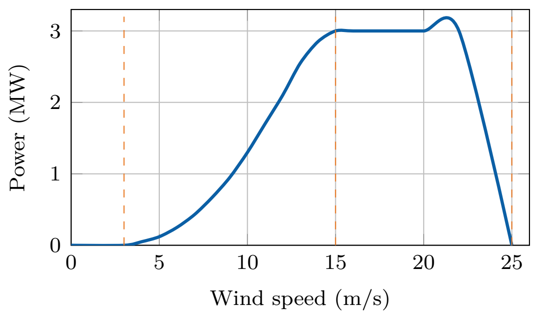
Key Figures

Rotor diameter	120–250 m
Rated power	3–18 MW
Cut-in speed	3–4 m/s
Rated speed	11–15 m/s
Cut-out speed	20–25 m/s
Typical capacity factor	30–60%

Wind-turbine architecture



Typical power curve



The cut-in, rated and cut-out regions determine how the turbine operates safely.

Engineering Insight

A larger rotor does more than increase rated power. It also captures more energy at moderate wind speeds, which can improve annual production. However, longer blades experience higher loads and are more difficult to manufacture, transport and recycle.

Useful Vocabulary

blade	pale
hub	moyeu
gearbox	multiplicateur
yaw	orientation de la nacelle
pitch	calage des pales
swept area	surface balayée
capacity factor	facteur de charge

Questions

1. Why does wind power depend on the cube of wind speed?
2. How does aerodynamic lift create rotor torque?
3. What is the purpose of pitch control?
4. What is the purpose of yaw control?
5. Interpret the main regions of the power curve.
6. Why can offshore wind achieve a higher capacity factor than onshore wind?

Engineering Challenge

A coastal region plans a 500 MW wind farm. Compare an onshore solution, a fixed-bottom offshore solution and a floating offshore solution. Consider energy yield, foundations, maintenance, grid connection, environmental impact and public acceptance. Use a weighted decision table and present a justified recommendation.

Application Exercise

A wind turbine has a rotor diameter of 100 m and operates in air of density 1.225 $\mathrm{kg\,m^{-3}}$ at a wind speed of 10 m/s. Using $P = \frac{1}{2}\rho A v^3 C_p$ with $C_p = 0.44$, estimate the aerodynamic power captured by the rotor. Compare C_p with the Betz limit, then estimate the electrical power delivered to the grid if the mechanical and electrical efficiency is 90%.

Speaking Task

Debate the statement: "Large wind farms should be built mainly offshore." Use technical, economic and environmental arguments, and respond to at least one opposing argument.

Sources and further reading

Global Wind Energy Council: online resource

IEA – Wind: online resource

EC04 – Nuclear Energy

Can nuclear power support a low-carbon future?

Learning Objectives

- Explain nuclear fission and a controlled chain reaction.
- Identify the energy and fluid flows in a pressurised water reactor (PWR).
- Explain why residual heat must be removed after shutdown.
- Interpret lifecycle-emission data and discuss the role of nuclear power.

Engineering News

Several countries are considering new reactors, lifetime extensions and Small Modular Reactors (SMRs). The main engineering questions concern construction time, financing, safety, fuel supply, waste management and operation within an electricity system containing more variable renewable generation.

Main Article

A nuclear power station converts energy released by **nuclear fission** into electricity. When a uranium-235 nucleus absorbs a neutron, it may split into two lighter nuclei. This reaction releases energy and additional neutrons, which can trigger further fissions. In a reactor, the number of fissions is controlled so that power remains stable.

In a pressurised water reactor, fuel assemblies are placed inside the reactor vessel. Water in the primary circuit flows through the core and removes heat. It is kept at high pressure so that it remains liquid at about 300 °C. In the steam generator, the primary water transfers heat to a physically separate secondary circuit.

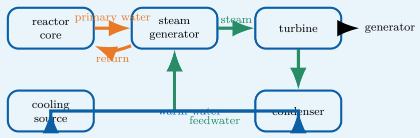
The secondary water boils and produces steam. The steam drives a turbine connected to an electrical generator. After the turbine, a condenser turns the steam back into liquid water. A third circuit carries waste heat towards a river, the sea or a cooling tower. The three circuits are separated because they have different functions and operating conditions.

Reactor power is adjusted by controlling neutron absorption. Control rods contain materials that absorb neutrons, while soluble boron can also be added to the primary water. Even after the chain reaction stops, radioactive decay continues to release **residual heat**. Cooling must therefore remain available after shutdown.

Safety is based on defence in depth: prevention, monitoring, automatic protection, backup systems and several physical barriers. Engineers also study component ageing, extreme weather, human factors, cybersecurity and long-term waste management.

Engineering Focus

Energy conversion in a PWR



The primary circuit transports nuclear heat, the secondary circuit produces steam, and the cooling circuit rejects unused heat. A PWR is therefore both a nuclear system and a thermal power plant.

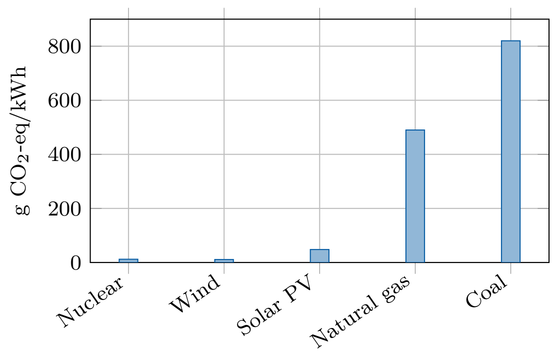
Did You Know?

Stopping fission does not stop heat production immediately. Residual heat decreases rapidly, but cooling remains essential for hours, days and longer after shutdown.

Key Figures

Typical French PWR output	900–1450 MW
Primary-water temperature	about 300 °C
Primary pressure	about 155 bar
Typical capacity factor	70–90%
Operating life	several decades

Indicative lifecycle emissions



Indicative median lifecycle values based on IPCC assessments. Actual values vary with technology, location and methodology.

Engineering Insight

A power station must remove more thermal energy than it converts into electricity. This is why condenser performance and access to a cooling source strongly influence plant design, site selection and output during heat waves.

Fission and fusion

Fission splits heavy nuclei and is used in current commercial reactors. **Fusion** combines light nuclei and powers the Sun. Fusion research is active, but fusion is not yet a commercial source of electricity.

Useful Vocabulary

chain reaction	réaction en chaîne
reactor vessel	cuve du réacteur
fuel assembly	assemblage combustible
control rod	barre de contrôle
steam generator	générateur de vapeur
containment	enceinte de confinement
residual heat	puissance résiduelle
spent fuel	combustible usé

Questions

1. What happens during nuclear fission?
2. Why must the primary water remain under high pressure?
3. Explain the function of each of the three water circuits.
4. How do control rods modify reactor power?
5. Why is cooling required after shutdown?
6. Use the graph to compare nuclear power with fossil-fuel generation.
7. Why can high air or river temperatures reduce the output of a power station?

Engineering Challenge

A region must replace 8 GW of fossil-fuelled generation while reducing lifecycle CO_2 emissions and maintaining supply during winter evenings. Propose a balanced mix using nuclear, wind, solar, storage, interconnections and demand response. Justify installed capacity, expected availability, flexibility, environmental constraints, construction sequence and major risks.

Application Exercise

A 1.3 GW nuclear unit operates with a capacity factor of 88%. Estimate its annual electrical production in TWh. Then determine how many 3 MW wind turbines operating at a 32% capacity factor would produce the same annual energy. Explain why equal annual production does not mean that the two systems provide electricity in the same way.

Speaking Task

Debate: "Europe should build new nuclear power stations." Use evidence about safety, climate, cost, waste, construction time and security of supply. Each team must answer one technical objection from the opposing side.

Sources and further reading

IAEA – Nuclear Power Reactors: online resource

OECD Nuclear Energy Agency: online resource

IPCC – Energy Systems: online resource

EC05 – Energy Storage

Why is energy storage essential to the energy transition?

Learning Objectives

- Explain why electrical grids need storage and flexibility.
- Distinguish power, stored energy and discharge duration.
- Compare storage technologies using efficiency, response time, lifetime and scale.
- Select a suitable storage mix for a given engineering problem.

Engineering News

As solar and wind generation expands, grid operators are deploying batteries, pumped-storage plants and other flexible resources. The objective is not to find one universal storage device, but to combine technologies operating over seconds, hours, days and seasons.

Did You Know?

A highly efficient storage system can still be unsuitable if its response time, discharge duration, lifetime, safety requirements or location do not match the required service.

Main Article

Electricity networks must keep production and consumption balanced at every instant. A sudden mismatch changes grid frequency and may damage equipment or trigger automatic disconnections. Storage systems absorb energy when production is greater than demand and return it when the system requires additional power.

Engineers distinguish **power** from **energy**. Power, measured in watts, describes how quickly a system can charge or discharge. Stored energy, measured in watt-hours or joules, describes how long that power can be maintained. Their ratio gives an approximate discharge duration:

$$t = \frac{E}{P}.$$

A 20 MW battery containing 40 MWh can therefore operate at full power for about two hours.

Lithium-ion batteries respond in fractions of a second and have a high round-trip efficiency. They are suitable for frequency control, electric vehicles and shifting solar energy from midday to the evening. Their limitations include ageing, fire protection, cost and the supply of raw materials.

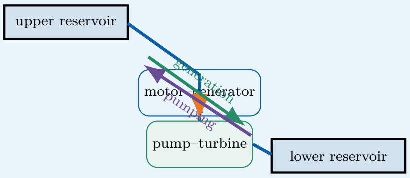
Pumped-storage hydroelectricity stores much larger amounts of energy. Water is pumped to an upper reservoir and later released through a reversible turbine. Such plants can operate for many decades, but they require suitable terrain and major civil-engineering works.

Compressed air, thermal storage and hydrogen can address longer durations. Hydrogen is produced by electrolysis, stored, and later used in a fuel cell or turbine. Its round-trip efficiency is lower because several energy conversions are required. It may nevertheless be useful when duration and capacity matter more than efficiency, especially for seasonal balancing.

No storage option is optimal for every task. A resilient low-carbon grid combines storage with interconnections, demand response, controllable generation and accurate forecasting.

Engineering Focus

Pumped-storage hydroelectricity

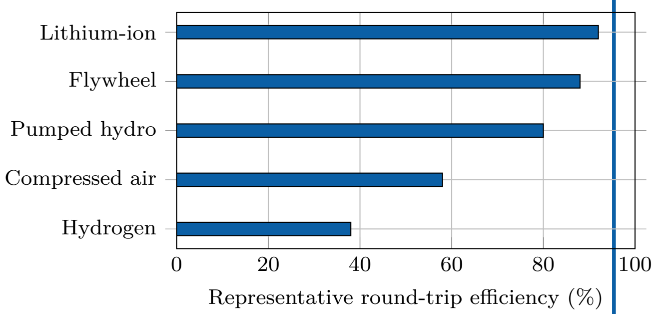


During low demand, electrical energy drives the pump and raises water. During high demand, the water flows downhill and drives the turbine. The stored gravitational energy is $E = mgh$.

Key Figures

Lithium-ion round-trip efficiency	85–95%
Pumped-storage hydro	70–85%
Hydrogen power-to-power	30–45%
Typical battery duration	1–8 h
Seasonal storage	weeks–months

Storage technologies compared



Representative values only: actual performance depends on design and operating conditions.

Engineering Insight

Storage sizing requires two independent decisions. The power rating must cover the largest instantaneous imbalance, while the energy capacity must cover the required duration. A system can have enough power but run empty too soon, or contain enough energy but deliver it too slowly.

Useful Vocabulary

round-trip efficiency	rendement aller-retour
charge / discharge	charge / décharge
storage duration	durée de stockage
reservoir	réservoir
flywheel	volant d'inertie
demand response	effacement de consommation
grid frequency	fréquence du réseau

Questions

1. Why must electricity production and consumption remain balanced?
2. Distinguish power from stored energy.
3. A 12 MW system stores 48 MWh. What is its full-power discharge duration?
4. Why are batteries suitable for rapid grid services?
5. Explain the operating cycle of a pumped-storage plant.
6. Why can hydrogen be useful despite its lower efficiency?
7. Which criteria should engineers consider besides efficiency?

Engineering Challenge

A remote island has a 12 MW peak demand, a 20 MW wind farm and a 10 MW solar farm. Propose a storage strategy for frequency control, night-time supply and a five-day period of weak wind. Select at least two technologies and justify their power, energy capacity, duration, efficiency, safety and environmental impact. Present a block diagram and a table of design choices.

Application Exercise

A pumped-storage plant transfers 5.0×10^6 m^3 of water through a vertical height of 420 m. Estimate the stored gravitational energy using $E = \rho V g h$, with $\rho = 1000\ \mathrm{kg\,m^{-3}}$. Express the result in GWh, then estimate the recoverable energy for a round-trip efficiency of 78%. Finally, calculate how long the plant could supply 1.0 GW at constant power.

Speaking Task

In teams, defend one storage technology in a three-minute technical pitch. Explain where it performs well, then identify one service for which it would be a poor choice.

Sources and further reading

IEA – Grid-scale storage: online resource

NREL – Energy storage: online resource

EC06 – Electric Vehicles

Are electric cars really sustainable?

Learning Objectives

- Describe the energy and information flows in a battery-electric vehicle.
- Explain charging, regenerative braking and battery thermal management.
- Evaluate an electric-mobility project using technical, economic and environmental criteria.

Engineering News

Electric mobility is moving beyond the vehicle itself. Engineers now design complete systems that connect cars, charging stations, buildings and electricity networks. Smart charging can shift demand away from peak hours, while vehicle-to-grid operation may allow parked vehicles to support the power system.

Engineering Insight

Increasing battery capacity improves range, but it also increases mass, cost, material demand and charging time. The best design is therefore not the vehicle with the largest battery, but the one whose capacity matches the real mission.

Main Article

A battery-electric vehicle stores electrical energy in a high-voltage battery pack. During acceleration, the battery supplies direct current to a power-electronics inverter. The inverter controls the alternating current delivered to the traction motor, which converts electrical energy into mechanical torque at the wheels. Unlike a combustion engine, an electric motor can deliver high torque from very low speed and usually maintains high efficiency over a wide operating range.

The battery pack is not simply a collection of cells. It also contains electrical connections, contactors, sensors, cooling channels, structural protection and a **Battery Management System** (BMS). The BMS estimates state of charge and state of health, balances the cells and protects the pack against excessive current, voltage and temperature. Thermal management is essential because low temperatures reduce available power, whereas high temperatures accelerate ageing and may increase safety risks.

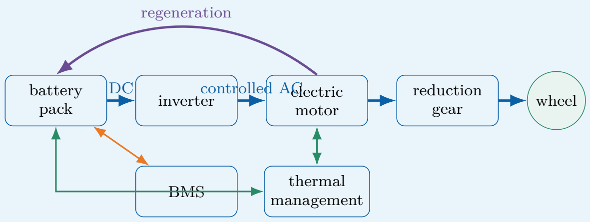
When the driver brakes, the inverter can operate the traction motor as a generator. Part of the vehicle's kinetic energy is then converted back into electrical energy and stored in the battery. This **regenerative braking** reduces energy consumption and mechanical-brake wear. Friction brakes nevertheless remain necessary for emergency stops, low-speed operation and situations in which the battery cannot accept additional power.

Charging power depends on both the charging station and the vehicle. An on-board charger converts alternating current from a domestic or public AC supply. Fast DC charging bypasses the on-board charger and feeds controlled direct current to the battery. High charging power shortens stops, but it also increases heat generation, infrastructure cost, battery stress and demand on the local electricity network.

Electric vehicles have no exhaust emissions during use, but they are not impact-free. Engineers must consider battery manufacture, electricity generation, vehicle mass, tyre particles, lifetime, repairability and recycling. Their environmental performance therefore depends on the complete life cycle and on the electricity mix used for charging.

Engineering Focus

Energy flow in a battery-electric vehicle

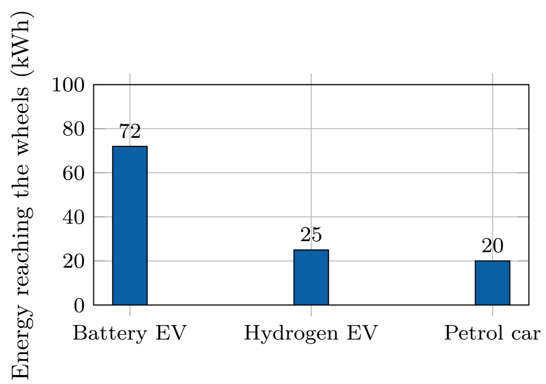


The high-voltage traction circuit is complemented by a DC/DC converter that supplies the low-voltage network. Isolation monitoring and contactors disconnect the battery if a serious fault is detected.

Key Figures

Electric motor peak efficiency	90–97%
Typical battery voltage	350–800 V
Home AC charging	2–11 kW
Rapid DC charging	50–350 kW
Typical passenger-car battery	40–100 kWh

From supplied energy to the wheels



Illustrative conversion chain starting from 100 kWh of supplied energy. Actual values vary with technology, driving conditions and system boundaries.

Useful Vocabulary

battery pack	pack batterie
state of charge	état de charge
state of health	état de santé
traction motor	moteur de traction
regenerative braking	freinage régénératif
on-board charger	chargeur embarqué
charging point	point de recharge
power electronics	électronique de puissance

Questions

1. What are the functions of the inverter and the traction motor?
2. Why does a battery pack require thermal management?
3. Explain how regenerative braking changes the direction of energy flow.
4. Compare AC charging with rapid DC charging.
5. Why can charging power be limited by the vehicle rather than the station?
6. Use the graph to compare the three energy pathways.
7. Why is the largest possible battery not always the best design choice?

Engineering Challenge

A city operates 24 buses, each travelling 220 km per day. Compare overnight depot charging with high-power opportunity charging at selected terminal stops. Propose a battery capacity, charging power and operating schedule. Build a decision table using range, charging time, battery ageing, infrastructure cost, network connection, redundancy, safety and maintenance. Conclude with a justified technical recommendation.

Application Exercise

An electric bus has a usable battery capacity of 320 kWh and consumes on average 1.25 kWh/km. Estimate its theoretical range. Then calculate the energy recovered during a 12 km downhill section if the mechanical energy available at the wheels is 18 kWh and the regenerative chain efficiency is 72%. Explain why the practical operating range must remain lower than the theoretical value.

Speaking Task

Prepare a four-minute technical briefing answering the question: “Should all new urban cars be electric by 2035?” Present one technical advantage, one limitation and one system-level measure that could improve urban mobility.

Sources

IEA – Global EV Outlook: online resource

U.S. Department of Energy – Electric vehicles: online resource

Alternative Fuels Data Center – How electric vehicles work: online resource

EC07 – Robotics

How can robots work safely with humans?

Learning Objectives

- Identify the mechanical, electrical, control and safety subsystems of a robot.
- Explain position, force and vision feedback in a collaborative workstation.
- Design and justify a safe robotic solution for an industrial task.

Engineering News

Collaborative robots, or **cobots**, are changing the organisation of factories and laboratories. Instead of operating permanently behind a fence, they can share selected tasks with people. The engineering objective is not merely to make the robot slower: the complete workstation must detect hazards, limit energy and bring the system to a safe state when required.

Engineering Insight

A cobot is not automatically a safe collaborative application. Safety depends on the robot, tool, workpiece, programmed speed, human task and surrounding equipment. The risk assessment must therefore cover the complete workstation.

Main Article

A robot is a programmable machine that senses its environment, processes information and acts through a mechanical structure. A typical articulated robot contains rigid links connected by joints. Electric servomotors create torque at these joints, while encoders measure angular position and speed. A controller compares the measured motion with the requested trajectory and continuously corrects the motor commands.

The tool attached to the final link is called the **end-effector**. It may be a gripper, a welding torch, a screwdriver or a camera. Selecting this tool requires engineers to consider payload, accuracy, cycle time, gripping force and product geometry. The useful region that the end-effector can reach is the robot's workspace.

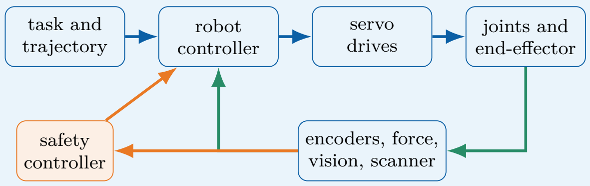
Position control is sufficient for many repetitive tasks, but physical interaction requires additional information. A force–torque sensor can detect contact, while motor-current monitoring can estimate resistance at a joint. Machine-vision systems locate parts, inspect surfaces and track people. These measurements form feedback loops that allow the robot to adapt rather than repeat a completely fixed sequence.

Safe collaboration depends on the entire application. Possible strategies include monitored stops, hand-guiding, speed and separation monitoring, and power-and-force limitation. A scanner or camera can define protective zones around the robot. As a person approaches, the controller first reduces speed and then stops hazardous motion. Mechanical design also matters: rounded surfaces, low moving mass and the avoidance of crushing points can reduce injury risk.

Mobile robots combine localisation, mapping and path planning. Because sensors can fail or be obstructed, their design requires diagnostics and a defined safe response. Cybersecurity is equally important: access control, validated software and network segmentation help prevent dangerous commands or corrupted sensor data.

Engineering Focus

Closed-loop architecture of a collaborative robot

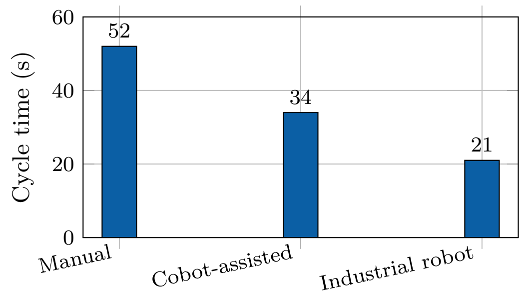


The standard motion controller aims to achieve the task. The safety controller independently supervises hazardous conditions and can request a controlled stop or remove drive power.

Key Figures

Typical articulated cobot axes	6–7
Repeatability of small industrial robots	about 0.02–0.1 mm
Typical cobot payload range	3–35 kg
Main feedback period	milliseconds
Emergency response objective	safe state

Assembly cycle comparison



Illustrative values for one assembly operation. A faster cycle does not by itself prove that a solution is safer, more flexible or more economical.

Useful Vocabulary

joint	articulation
link	solide / bras
actuator	actionneur
encoder	codeur
end-effector	effecteur terminal
payload	charge utile
repeatability	répétabilité
protective stop	arrêt de protection

Questions

1. What information is provided by an encoder?
2. Distinguish accuracy from repeatability.
3. Why may position control be insufficient during contact tasks?
4. Explain speed and separation monitoring.
5. Why must the end-effector be included in the risk assessment?
6. Give two possible consequences of a cybersecurity failure.
7. Use the graph to explain why productivity cannot be the only selection criterion.

Engineering Challenge

A company wants to automate the insertion and tightening of four screws on a 2.5 kg product while an operator performs the adjacent wiring task. Design the collaborative workstation. Select the robot, end-effector and sensors; define the work sequence and protective zones; identify crushing, impact, tool and cybersecurity hazards; and propose validation tests. Compare the solution with a fully manual station and a fenced industrial robot using a weighted decision matrix.

Application Exercise

A collaborative robot moves a 4.0 kg component with a horizontal acceleration of 1.8 m/s^2. Calculate the resulting inertial force. If the end-effector has a mass of 1.5 kg, repeat the calculation for the complete moving load. Explain why this simple calculation is not sufficient to validate human–robot safety.

Speaking Task

Prepare a four-minute design review answering the question: “Should this workstation use a cobot or a conventional industrial robot?” Defend your choice using safety, cycle time, flexibility, payload, maintenance and cost.

Sources

International Federation of Robotics – industrial and service robots: online resource

NIST – robotics and autonomous systems: online resource

European Commission – machinery safety: online resource

EC08 – Artificial Intelligence

Can machines make reliable engineering decisions?

Learning Objectives

- Distinguish rule-based programming, machine learning, training and inference.
- Explain how dataset quality and thresholds affect classification performance.
- Design and justify a validated AI architecture for an engineering application.

Engineering News

Artificial intelligence is now embedded in industrial inspection, predictive maintenance, energy management, medical imaging and autonomous systems. Its engineering value comes from its ability to process large quantities of sensor data, but performance figures alone are not enough: reliability, traceability and human responsibility must also be designed.

Engineering Insight

A model can achieve high overall accuracy while still missing most rare defects. Engineers must therefore select performance indicators from the consequences of errors, not from a single headline score.

Main Article

Traditional software follows explicit rules written by a programmer. A machine-learning system instead learns a mathematical model from examples. During **training**, an algorithm adjusts internal parameters so that the model's outputs approach the expected answers contained in a labelled dataset. During **inference**, the trained model processes new data and produces a classification, numerical estimate or recommended action.

In computer vision, a neural network can learn to distinguish acceptable products from defective ones. The input may be an image of a machined surface, while the output is a defect category and a confidence score. In predictive maintenance, vibration, temperature and current measurements can be used to estimate the condition of a bearing or motor. In both cases, the model is only one part of the complete engineering system: sensors, lighting, communication, computing hardware and the human–machine interface also determine the final result.

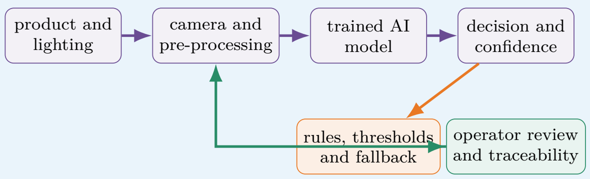
Dataset quality is critical. If defects are rare, poorly labelled or photographed under only one lighting condition, the model may appear accurate while failing in production. Engineers therefore separate data into training, validation and test sets. The test set must remain independent so that it measures generalisation rather than memorisation. Useful indicators include precision, recall, false-alarm rate and missed-detection rate.

Safety-critical applications require additional safeguards. A confidence threshold can prevent uncertain outputs from triggering an automatic decision. Independent physical checks, plausibility rules and human confirmation can also reduce risk. Engineers must monitor **data drift**: over time, new products, sensor ageing or environmental changes may make operating data different from the original training data.

Explainability and traceability are also important. The development team should record dataset versions, model parameters, test results and software updates. If the system influences a hazardous action, responsibility remains with the organisation that designed and validated the complete system, not with the algorithm itself.

Engineering Focus

Validated AI inspection architecture

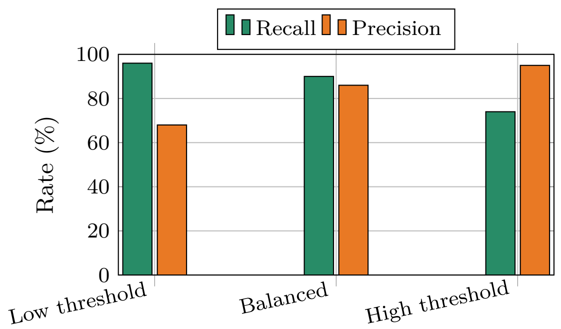


The AI model proposes a result. Independent thresholds and fallback rules determine whether it may be accepted automatically or must be reviewed by a person. Production data and operator feedback are stored so that performance can be monitored over time.

Key Figures

Training set	model fitting
Validation set	parameter selection
Test set	independent evaluation
False negative	missed defect
False positive	false alarm
Confidence threshold	application-dependent

Defect-detection trade-off



Illustrative values. Lowering the threshold detects more defects but may also create more false alarms. The correct setting depends on the consequences of each error.

Useful Vocabulary

dataset	jeu de données
training	entraînement
inference	inférence
label	étiquette / classe
accuracy	exactitude globale
precision	précision positive
recall	taux de détection
data drift	dérive des données

Questions

1. Distinguish training from inference.
2. Why must a test set remain independent?
3. Explain the difference between a false positive and a false negative.
4. Why can overall accuracy be misleading for rare defects?
5. Use the graph to explain the threshold trade-off.
6. Give two possible causes of data drift.
7. Why must responsibility remain with the engineering organisation?

Engineering Challenge

A factory wants to inspect 30 aluminium parts per minute using a camera and an AI model. A missed crack may lead to a dangerous failure, while a false alarm sends a good part for manual inspection. Design the complete inspection station. Specify the camera and lighting arrangement, dataset, performance indicators, acceptance threshold, fallback strategy, traceability records and validation tests. Include a monitoring plan for data drift and define the conditions that require human review or model retraining.

Application Exercise

An image-inspection system processes 240 parts per minute. Its false-rejection rate is 1.5% and its false-acceptance rate is 0.20%. During an eight-hour shift, estimate the number of acceptable parts wrongly rejected if 96% of production is actually conforming, and the number of defective parts wrongly accepted. State one operational consequence of each type of error.

Speaking Task

Prepare a four-minute design review answering the question: “Should this inspection system reject parts automatically?” Defend your position using safety, productivity, precision, recall, traceability and human supervision.

Sources

NIST – Artificial Intelligence: online resource
OECD – Artificial Intelligence: online resource
European Commission – Artificial Intelligence: online resource

EC09 – Space Engineering

How do engineers design systems for extreme environments?

Learning Objectives

- Identify the main subsystems and interfaces of an artificial satellite.
- Explain how engineers manage launch loads, vacuum, radiation and temperature variations.
- Build and justify a coherent mass, power and reliability budget for a small mission.

Engineering News

Reusable launch vehicles, commercial constellations and small satellites are changing access to space. Lower launch costs make new missions possible, but every spacecraft must still survive an extreme environment and operate for years without direct maintenance.

Main Article

A spacecraft is designed around its **mission**. An Earth-observation satellite must carry a camera or radar payload, point it accurately towards the ground, store the collected data and send it to a ground station. Around this payload, engineers build a service platform containing the structure, power supply, thermal control, communication system, onboard computer and attitude-control system.

Launch is the first major challenge. The satellite experiences acceleration, vibration and intense acoustic pressure. Its structure must therefore be light but stiff, while sensitive components are mounted and tested so that they do not resonate dangerously. After separation from the launcher, the environment changes completely: there is almost no atmosphere, heat cannot be removed by convection and electronic components are exposed to radiation.

Electrical power usually comes from photovoltaic panels. Batteries supply the spacecraft during eclipse, when Earth blocks the Sun. A power-control unit regulates voltage and distributes energy to the payload, computer, heaters and transmitters. Because available power is limited, mission operations are planned carefully: a high-power transmitter or instrument may only be activated at specific times.

Thermal control is equally important. In sunlight, exposed surfaces may become very hot; in shadow, they may cool rapidly. Engineers use insulation blankets, reflective coatings, heat pipes, radiators and electrical heaters. The objective is not to keep every part at room temperature, but to keep each component within its qualified operating range.

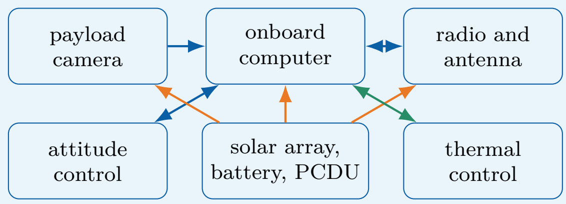
A satellite must also control its orientation, called **attitude**. Sun sensors, star trackers and gyroscopes estimate its orientation. Reaction wheels rotate internally and make the spacecraft turn in the opposite direction. Small thrusters can desaturate the wheels or perform orbital corrections. Since repair is generally impossible, critical functions may be duplicated and the onboard software must detect faults and switch to a safe mode.

Engineering Insight

Every additional subsystem improves capability or reliability, but also consumes mass, volume and power. Spacecraft design is therefore governed by tightly coupled budgets rather than by independent component choices.

Engineering Focus

Typical small-satellite architecture

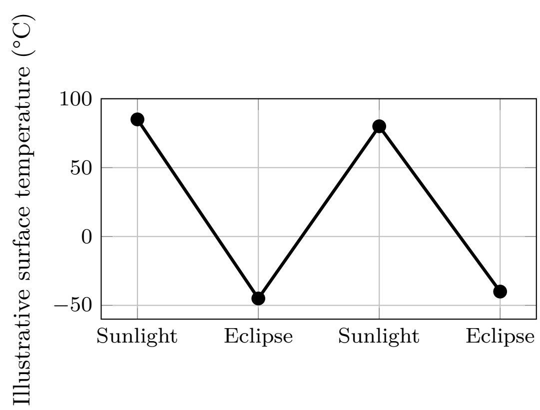


The payload performs the mission, while the platform keeps it powered, pointed, protected and connected. Interfaces between subsystems are as important as the components themselves.

Key Figures

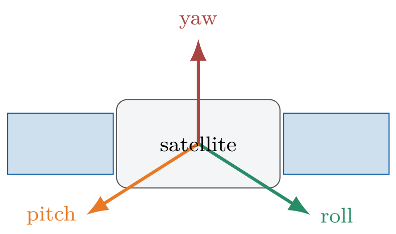
Typical low Earth orbit altitude	400–800 km
Orbital speed	about 7.8 km/s
One orbit	about 90 min
Convection in vacuum	unavailable
Repair after launch	usually impossible

Thermal cycle in orbit



Illustrative values only. Real temperatures depend on orbit, surface coating, orientation, internal dissipation and thermal-control design.

Three-axis attitude control



Sensors estimate orientation; reaction wheels or thrusters create the required rotation around the three axes.

Useful Vocabulary

spacecraft	engin spatial
payload	charge utile
launch load	effort au lancement
solar array	panneau solaire
eclipse	passage dans l'ombre
attitude	orientation
reaction wheel	roue de réaction
star tracker	viseur d'étoiles
thruster	propulseur
redundancy	redondance

Questions

1. Which mechanical loads affect a satellite during launch?
2. Why is convection unavailable in space?
3. What is the role of the battery during eclipse?
4. Name three devices used for attitude determination or control.
5. Why are critical functions sometimes duplicated?
6. Use the thermal graph to identify the main thermal-control challenge.
7. Why must mass, power and reliability budgets be considered together?

Engineering Challenge

Design a 20 kg Earth-observation satellite. Draw a subsystem diagram and prepare preliminary mass and power budgets. Explain how the satellite will store energy, control temperature, point its camera, communicate with Earth and react to one critical failure. Identify at least two design compromises involving mass, power, cost, image quality or reliability, then justify the final architecture.

Application Exercise

A satellite in low Earth orbit receives 950 W from its solar arrays in sunlight and requires an average electrical power of 620 W. During a 36-minute eclipse, the battery supplies the complete load. Estimate the minimum energy that must be delivered by the battery in kWh. Then calculate the required nominal capacity if only 70% of the battery capacity may be used.

Speaking Task

Present your satellite in a four-minute mission review. Explain the mission, architecture, budgets and most serious technical risk. Conclude by defending one design compromise.

Sources

NASA Small Spacecraft Systems Virtual Institute: online resource

European Space Agency – Space Engineering and Technology: online resource

EC10 – Medical Engineering

How can technology improve healthcare?

Learning Objectives

- Describe the complete measurement and decision chain of a connected medical device.
- Explain the roles of calibration, filtering, biocompatibility and risk analysis.
- Design and justify a safe monitoring system combining sensors, software and human supervision.

Engineering News

Wearable sensors, robotic surgery, smart prostheses and remote monitoring are transforming healthcare. These systems can improve diagnosis and quality of life, but they must meet demanding requirements for safety, reliability, privacy and clinical usefulness.

Main Article

Medical engineering combines electronics, mechanics, materials science, computer science and human physiology. A medical device may simply measure a quantity, such as body temperature, or it may act on the patient, as a pacemaker, insulin pump or powered prosthesis does. In every case, design must begin with a clearly defined medical need and a careful analysis of possible hazards.

Most monitoring devices contain a measurement chain. A sensor converts a physiological quantity into an electrical signal. Because this signal is often weak and noisy, analogue electronics amplify and filter it before an analogue-to-digital converter sends numerical data to a processor. Software then calculates indicators, detects abnormal values and displays information to a patient or healthcare professional. Calibration links the electrical output to a trustworthy physical measurement.

A pulse oximeter, for example, shines red and infrared light through a finger and measures how much light is absorbed. From the changing optical signals, the device estimates blood oxygen saturation and pulse rate. Movement, poor contact or ambient light can disturb the measurement, so the system must detect low-quality signals rather than display a misleading result.

Implantable devices create additional constraints. Their materials must be biocompatible, their energy consumption must be extremely low and their enclosure must protect the electronics from body fluids. A pacemaker continuously monitors cardiac activity and delivers a controlled electrical pulse only when required. This is a closed-loop system: measurement, decision and action are linked.

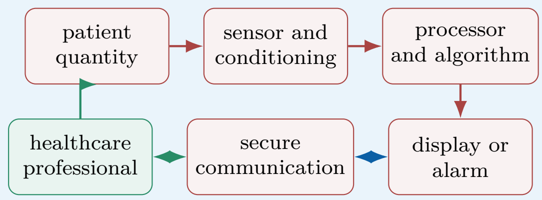
Connected devices can transmit measurements to a smartphone or hospital server. Remote monitoring may allow earlier intervention, but it also introduces cybersecurity and privacy risks. Engineers must protect communication, control software updates and ensure that a network failure cannot place the patient in danger. Clinical validation remains essential: a technically accurate sensor is useful only if its information supports an appropriate medical decision.

Engineering Insight

A device can be precise without being accurate. Repeated measurements may be very similar but all shifted away from the true value. Calibration identifies systematic error, while validation determines whether the complete device is suitable for its intended clinical use.

Engineering Focus

Connected patient-monitoring chain

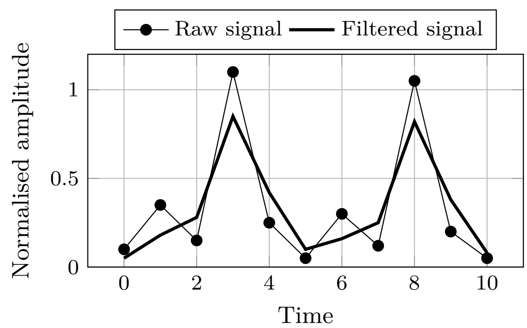


The device provides information, but the complete system includes the patient, user interface, communication network and medical decision. Each interface must be validated.

Key Figures

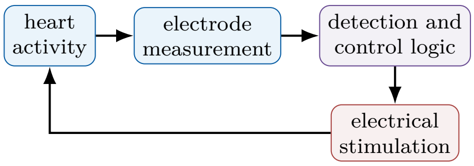
Sensor output	often millivolts or microvolts
Implant operating life	several years
Acceptable false alarm rate	application-dependent
Software traceability	throughout development
Patient data	confidential

Filtering a physiological signal



Illustrative data. Filtering reduces rapid variations, but excessive filtering may delay or distort clinically important information.

Closed-loop pacemaker principle



The controller stimulates the heart only when the measured rhythm does not satisfy the programmed criteria. A safe mode must remain available if a sensor or software fault is detected.

Useful Vocabulary

medical device	dispositif médical
implant	implant
prosthesis	prothèse
biocompatible	biocompatible
calibration	étalonnage
signal conditioning	conditionnement du signal
false alarm	fausse alarme
clinical validation	validation clinique
privacy	confidentialité
closed-loop control	commande en boucle fermée

Questions

1. Why must a weak sensor signal be amplified and filtered?
2. Distinguish precision, accuracy and clinical validity.
3. Give two possible disturbances affecting a pulse oximeter.
4. Why must an implant consume very little energy?
5. What makes a pacemaker a closed-loop system?
6. Use the graph to explain one benefit and one risk of filtering.
7. Why must a network failure not disable essential medical functions?

Engineering Challenge

Design a connected device that monitors an elderly patient at home. Select two physiological or activity measurements, draw the information chain and define alarm conditions. Explain how the system detects poor-quality data, limits false alarms, protects privacy, continues operating during a network failure and keeps a healthcare professional in the decision loop. Add a verification and validation plan covering calibration, software, cybersecurity and representative user tests.

Application Exercise

A physiological sensor produces 2.4 mV for the measured condition. The acquisition system requires a 1.8 V signal at the analogue-to-digital converter input. Calculate the ideal amplifier gain. If an offset of 15 mV is also amplified, calculate its contribution at the output and explain why offset correction is necessary.

Speaking Task

Should a connected medical device automatically contact emergency services when it detects a critical event? Present the benefits, risks and safeguards in a four-minute design review.

Sources

World Health Organization – Medical Devices: online resource

U.S. Food and Drug Administration – Medical Devices: online resource

EC11 – Additive Manufacturing

Can 3D printing transform industrial production?

Learning Objectives

- Explain the complete digital chain from CAD model to qualified part.
- Compare additive manufacturing with machining and casting using engineering criteria.
- Justify design, orientation, post-processing and inspection choices for a printed component.

Engineering News

Additive manufacturing is now used for rocket engines, aircraft components, medical implants and industrial tooling. Its real value is not simply the ability to print a shape, but the possibility of redesigning a function, reducing assembly operations and producing geometries that conventional processes cannot manufacture economically.

Main Article

Additive manufacturing creates a component layer by layer from a digital model. The process begins with a three-dimensional CAD file. Slicing software divides the geometry into thin horizontal sections and generates machine instructions. Depending on the technology, a nozzle deposits a molten polymer, a laser melts metal powder, or light solidifies a liquid resin.

This manufacturing route changes the designer's freedom. Internal cooling channels, lattice structures and organic shapes can be integrated into a single component. Topology optimisation can remove material from low-stress zones while preserving principal load paths. In aerospace engineering, this can reduce mass and fuel consumption. In medicine, a porous implant can be adapted to a patient and encourage bone growth.

The process also has important constraints. Overhanging zones may require temporary supports. These supports consume material, conduct heat and must be removed after printing. Thermal gradients can create residual stress, warping or cracks, especially in metal parts. Build orientation therefore affects support volume, surface quality, printing time, dimensional accuracy and mechanical properties.

A printed component is rarely ready for immediate use. Powder removal, heat treatment, support removal, machining and inspection may be required. Non-destructive testing can detect internal porosity, while dimensional measurement verifies that the geometry remains within tolerance. Since mechanical properties may vary with build direction, qualification is essential for safety-critical parts.

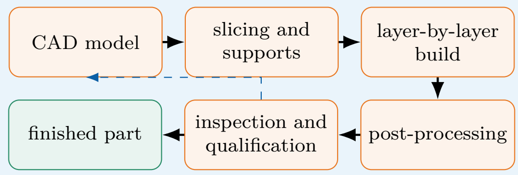
Additive manufacturing is particularly effective for prototypes, customised products, spare parts and low-volume high-value components. It is less competitive for millions of simple identical parts. The best industrial solution is often hybrid: printing creates the complex near-net shape, then conventional machining produces accurate interfaces and smooth functional surfaces.

Engineering Insight

A successful printed part is designed together with its manufacturing plan. Geometry, orientation, supports, scan strategy, heat treatment, machining references and inspection method form one integrated engineering decision.

Engineering Focus

Digital manufacturing chain

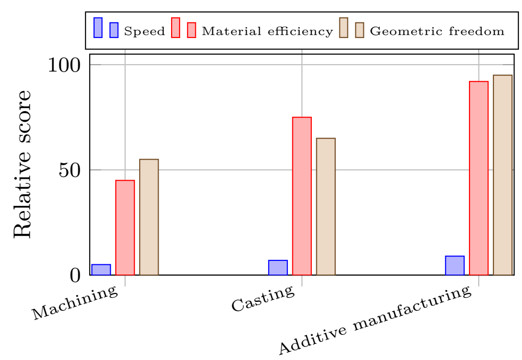


Inspection may lead to a design or parameter change. The chain is therefore iterative rather than strictly linear.

Key Figures

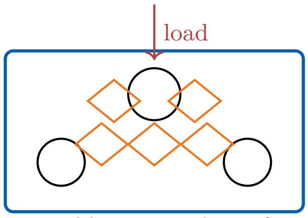
Typical polymer layer	0.10–0.30 mm
Typical metal layer	0.02–0.08 mm
Main digital input	3D CAD model
Frequent post-processes	heat treatment, machining
Best production range	low volume / high value

Comparing manufacturing routes



Illustrative comparison. The most suitable process depends on quantity, material, tolerance, geometry, qualification effort and total lifecycle cost.

Designing a lightweight bracket



material kept around interfaces and principal load paths

Topology optimisation must be followed by checks on minimum wall thickness, support removal, powder evacuation and machinable reference surfaces.

Useful Vocabulary

additive manufacturing	fabrication additive
slicing	découpage en couches
powder bed	lit de poudre
nozzle	buse
support structure	structure de support
build orientation	orientation de fabrication
warping	déformation / gauchissement
porosity	porosité
post-processing	post-traitement
topology optimisation	optimisation topologique

Questions

1. What information does slicing software generate?
2. Why does build orientation influence support volume and mechanical performance?
3. Give three examples of post-processing or inspection operations.
4. Why can mechanical properties be anisotropic?
5. Use the comparison chart to identify one advantage and one limitation of each process.
6. Why may part consolidation improve reliability but reduce repairability?

Application Exercise

A metal part is 120 mm high and is printed with a layer thickness of 0.04 mm. Each layer takes 18 seconds to scan and recoating adds no extra time in this simplified model.

1. Calculate the number of layers.
2. Estimate the ideal build time in hours.
3. Add 25% for calibration and interruptions. Estimate the total production time.
4. Explain why the real industrial lead time also includes preparation, cooling and inspection.

Engineering Challenge

Redesign an aluminium bracket for metal additive manufacturing. Define loads, interfaces and acceptance criteria; propose a lighter geometry; select a build orientation; estimate support and post-processing needs; identify inspection methods; and compare the total production route with machining from solid. Conclude with a justified recommendation.

Speaking Task

Will additive manufacturing replace conventional factories? Give a balanced four-minute answer with examples of suitable and unsuitable products, and distinguish technical possibility from economic relevance.

Sources

NIST – Additive Manufacturing: online resource

NASA – Additive Manufacturing: online resource

EC12 – Smart Materials

Can materials react to their environment?

Learning Objectives

- Identify the stimulus, physical coupling and response of several smart materials.
- Explain how shape-memory and piezoelectric effects are integrated into controlled systems.
- Evaluate stroke, force, response time, fatigue, safety and lifecycle constraints.

Engineering News

Smart materials can sense or respond to temperature, stress, electricity, light or magnetic fields. They are used in medical devices, vibration control, precision positioning and energy-efficient buildings, where one compact element can perform structural, sensing and actuation functions.

Main Article

A conventional material is selected mainly for passive properties such as stiffness, strength or thermal conductivity. A smart material produces a predictable response to an external stimulus. The response may be a change in shape, electrical charge, colour, viscosity or magnetic behaviour. Engineers use this physical coupling to create sensors, actuators and adaptive structures.

Shape-memory alloys are a well-known example. Nickel-titanium can be deformed in a low-temperature phase and recover a previously defined shape when heated. A thin wire can therefore act as a compact actuator: electrical current heats it, the wire contracts, and a spring or external load returns it when it cools. The available stroke is useful, but cooling may limit the operating frequency.

Piezoelectric materials couple mechanical and electrical energy. When compressed or bent, they produce electrical charge; this direct effect is used in force, pressure and vibration sensors. When voltage is applied, they deform slightly; this inverse effect is used in ultrasound transducers, fuel injectors and nanometre-scale positioning systems. Their motion is small, but rapid and precise.

Other smart materials provide different functions. Electrochromic layers change optical transmission when a voltage is applied. Magnetorheological fluids become more resistant to flow in a magnetic field and can adjust suspension damping. Self-healing polymers may release a repair agent when a crack ruptures embedded microcapsules.

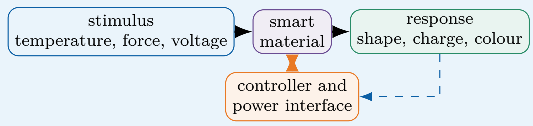
These materials are not intelligent by themselves. A useful engineering system still requires sensors, power electronics, control logic and safe limits. Performance changes with temperature, fatigue, hysteresis and ageing. Designers must consider response time, energy consumption, recyclability and failure behaviour. A passive safe state is often essential.

Engineering Insight

The material law alone does not define actuator performance. The useful output depends on geometry, preload, heat transfer, drive electronics, feedback control and the mechanical load connected to the material.

Engineering Focus

Stimulus–material–response chain

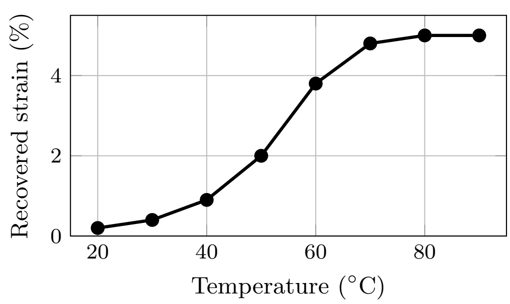


Closed-loop control measures the actual response and adjusts the stimulus. This improves accuracy and compensates for temperature variation, hysteresis and ageing.

Key Figures

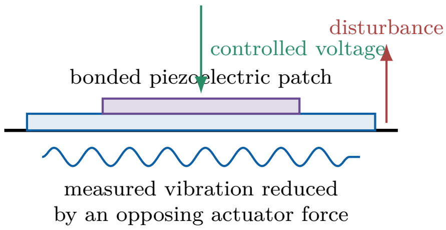
Shape-memory alloy strain	several percent
Piezoelectric actuator motion	micrometres
Piezoelectric response	very fast
Electrochromic transition	seconds to minutes
Critical design issues	fatigue, hysteresis, ageing

Shape recovery with temperature



Illustrative response of a shape-memory actuator. The transformation occurs over a temperature range rather than at one exact temperature.

Piezoelectric vibration control



A sensor measures vibration, the controller calculates a corrective command and the patch produces a small deformation that opposes the motion.

Useful Vocabulary

smart material	matériau intelligent
stimulus	sollicitation / stimulus
shape-memory alloy	alliage à mémoire de forme
phase transformation	transformation de phase
piezoelectric effect	effet piézoélectrique
strain	déformation relative
damping	amortissement
hysteresis	hystérésis
ageing	vieillissement
fail-safe state	état sûr en cas de panne

Questions

1. What distinguishes a smart material from a passive material?
2. How can a shape-memory wire act as an actuator?
3. Explain the direct and inverse piezoelectric effects.
4. Why is closed-loop control useful with smart materials?
5. Use the graph to estimate the useful transformation temperature range.
6. Why must fatigue, hysteresis and failure behaviour be considered together?

Application Exercise

A piezoelectric actuator is 40 mm long and produces a free strain of 0.08% when powered.

1. Calculate its free displacement in millimetres and micrometres.
2. A mechanical amplifier multiplies the displacement by 4. Estimate the output movement.
3. If the loaded displacement is only 65% of the free value, calculate the useful movement.
4. Explain why amplification usually reduces available force or stiffness.

Engineering Challenge

Design an adaptive bicycle-helmet ventilation system using a smart material. Define requirements, stimulus, response, actuator geometry and control strategy. Estimate the required movement, compare at least two candidate materials, identify fatigue and thermal risks, and explain how the helmet remains safe if the active material or power supply fails.

Speaking Task

Choose one smart material and present an everyday product that could use it. Explain the benefit, technical limitation, control requirement and environmental impact in a four-minute product review.

Sources

NIST – Materials Research: online resource

European Space Agency – Space Engineering and Technology: online resource

EC13 – Smart Cities

Can connected systems make cities more sustainable?

Learning Objectives

- Describe the architecture and control loop of a connected urban service.
- Quantify how sensing, communication and control can reduce resource consumption.
- Justify choices involving interoperability, privacy, resilience and social inclusion.

Engineering News

Cities are installing connected lighting, traffic and environmental-monitoring systems. The most successful projects do not simply collect data: they convert selected measurements into useful local decisions while protecting citizens and keeping essential services available during failures.

Main Article

A smart city uses sensors, communication networks and software to improve urban services. Distributed devices measure traffic flow, air quality, noise, temperature, water level or electricity consumption. Data are transmitted to a local controller or platform, where algorithms detect trends, identify abnormal conditions and select an appropriate action.

Street lighting provides a simple example. An adaptive system can combine a light sensor, motion detector and timetable. Lamps dim when streets are empty and return to a safe level when activity is detected. Energy savings are possible, but minimum illumination, response time and failure behaviour must be specified and verified.

Traffic management is more complex. Cameras, induction loops and connected vehicles estimate queue lengths. A controller can modify signal timing to reduce waiting time and emissions. Local optimisation of one junction, however, may create congestion elsewhere. Engineers must therefore model the complete network and define priorities for buses, emergency vehicles, cyclists and pedestrians.

Smart buildings and grids can cooperate. Occupancy sensors adjust heating and ventilation, while a district energy manager coordinates photovoltaic generation, batteries and electric-vehicle charging. This requires devices from different manufacturers to exchange data. Interoperability and open standards are essential. Edge computing can process urgent data near the sensors, reducing latency and dependence on a distant platform.

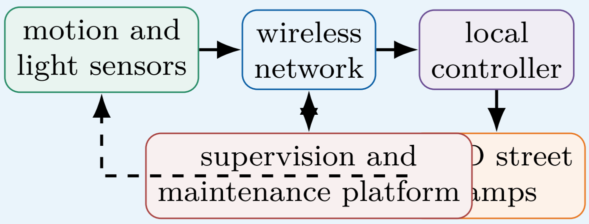
Connectivity also creates risks. Location or camera data may reveal personal behaviour. A network failure must not stop traffic lights, water pumps or emergency systems. Good design separates critical functions, minimises stored data, applies cybersecurity measures and provides redundant local fallback modes. A city is not truly smart if services remain inaccessible to people without smartphones or digital skills.

Engineering Insight

The value of an urban dataset depends on the decision it supports. Collecting more data increases communication, storage, cybersecurity and privacy costs; engineers should measure only what is needed to operate, maintain and verify the service.

Engineering Focus

Architecture of an adaptive lighting service

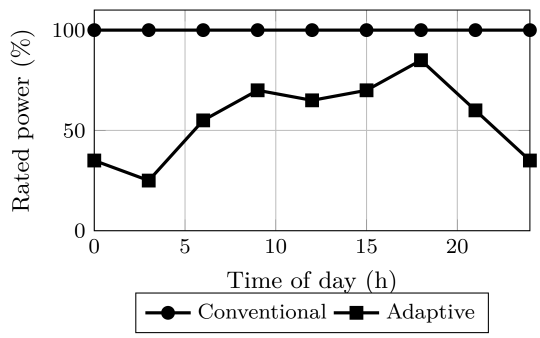


The essential control loop is local: sensing, decision and lighting remain available even if the central platform is disconnected.

Key Figures

Connected devices	thousands per district
Useful lamp response time	a few seconds
Important design property	interoperability
Critical service requirement	local fallback
Main ethical concern	privacy

Lighting power profile



Illustrative daily profiles. The adaptive system lowers demand during quiet periods while restoring a higher level when activity is detected.

Useful Vocabulary

sensor network	réseau de capteurs
traffic flow	flux de circulation
street lighting	éclairage public
occupancy	occupation / présence
interoperability	interopérabilité
digital twin	jumeau numérique
fallback mode	mode de repli
outage	panne / interruption
privacy	vie privée
urban planning	urbanisme

Questions

1. Describe the information and energy paths in the lighting system.
2. How can adaptive lighting save energy without reducing safety?
3. Why can local traffic optimisation be insufficient?
4. What does interoperability mean in this context?
5. Use the graph to identify the periods with the greatest potential savings.
6. Give three ways to make a connected urban service resilient.
7. Why should data minimisation be treated as an engineering requirement?

Application Exercise

A district operates 1,200 LED street lamps rated at 80 W. The conventional system runs at full power for 12 hours each night. An adaptive strategy uses full power for 5 hours and 35% power for the remaining 7 hours.

1. Calculate the nightly energy consumption of both systems.
2. Determine the energy saved per night and the percentage reduction.
3. Estimate the annual saving over 365 nights.
4. State two effects not represented by this simplified energy calculation.

Engineering Challenge

Design a smart energy-management system for a school district. Define requirements and performance indicators; select sensors, communication links and controlled equipment; draw the information flow; propose energy-saving rules; and describe local fallback modes. Include cybersecurity, maintenance, data minimisation and access for users without smartphones. Present a justified architecture rather than a list of connected devices.

Speaking Task

Should a city use cameras and connected sensors to reduce congestion? Present a four-minute design review covering one measurable benefit, one privacy risk, one resilience requirement and one condition for public acceptance.

Sources

European Commission – Smart Cities: online resource

NIST – Smart Connected Systems: online resource

EC14 – Water Treatment

How can engineering provide safe drinking water?

Learning Objectives

- Identify the principal barriers in a drinking-water treatment chain.
- Explain filtration, disinfection and reverse osmosis using physical principles.
- Select and monitor a robust treatment solution from water-quality and local constraints.

Engineering News

Membrane filtration, ultraviolet disinfection and lower-energy desalination are improving access to safe water. Yet the best solution is not always the most advanced one: reliability, operator skills, spare parts, energy availability and safe sludge or brine management often determine whether a plant succeeds over its complete lifetime.

Main Article

Raw water may contain suspended particles, microorganisms, organic matter, dissolved salts and chemical pollutants. No single process removes every contaminant, so drinking-water plants combine several barriers. The treatment chain is selected according to source-water quality, local constraints and the required quality of the final water.

Screens first remove leaves and large debris. During coagulation, a chemical reagent destabilises very small particles that would otherwise remain in suspension. Gentle mixing then creates larger flocs. In a sedimentation tank, these flocs settle under gravity and form sludge that must be collected, dewatered and treated.

Filtration removes the particles that remain. Sand filters trap material inside a porous bed, whereas activated carbon also adsorbs some organic molecules responsible for taste, odour or pollution. Membrane systems use pores of controlled size. As a filter becomes dirty, pressure loss increases, so periodic backwashing or chemical cleaning is necessary.

Disinfection inactivates harmful microorganisms. Chlorine can maintain protection throughout the distribution network, whereas ultraviolet light acts rapidly but leaves no disinfectant residual. Engineers often combine methods and monitor turbidity, pH, conductivity, flow and disinfectant concentration continuously.

Reverse osmosis is used for desalination. Pumps apply a pressure greater than the natural osmotic pressure, forcing water through a semi-permeable membrane while most salts remain in a concentrated brine. The process produces high-quality water but requires energy, pretreatment and careful brine management.

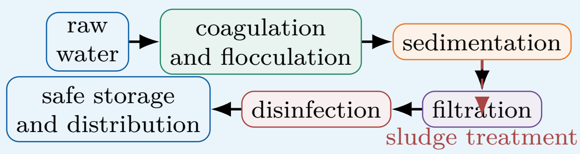
For a remote community, maintainability may be more important than maximum theoretical performance. Gravity flow, modular filters, solar pumping and simple sensors can reduce operating costs. A robust system must also include safe storage and prevent treated water from being contaminated again.

Engineering Insight

Safe water depends on verified barriers, not on appearance. A clear sample may still contain pathogens, while a highly sophisticated plant may fail if reagents, spare parts or trained operators are unavailable. Treatment performance and maintainability must therefore be designed together.

Engineering Focus

Multi-barrier treatment chain

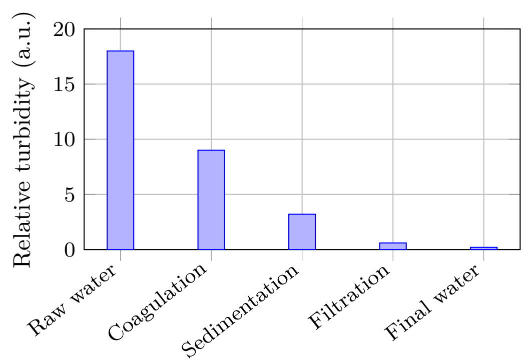


Each stage is a barrier. If one process performs less effectively, the remaining barriers reduce the risk of unsafe water reaching consumers.

Key Figures

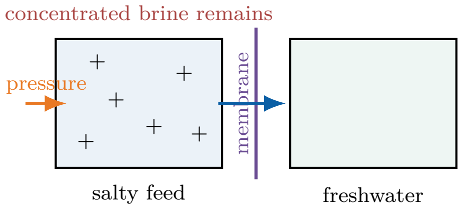
Raw-water quality	highly variable
Filtered-water target	very low turbidity
Reverse-osmosis driving force	high pressure
Main membrane issue	fouling
Essential final barrier	safe storage

Turbidity through the treatment chain



Successive barriers progressively reduce suspended matter. The values are illustrative; actual performance depends on source water and operating conditions.

Reverse osmosis principle



Useful Vocabulary

raw water	eau brute
coagulation	coagulation
floc	floc
sedimentation	décantation
activated carbon	charbon actif
backwashing	lavage à contre-courant
disinfection	désinfection
turbidity	turbidité
reverse osmosis	osmose inverse
brine	saumure

Questions

1. Why does a plant use several treatment barriers?
2. Explain the difference between coagulation and sedimentation.
3. What additional function can activated carbon provide?
4. Compare chlorine and ultraviolet disinfection.
5. Use the graph to identify the stage with the largest apparent turbidity reduction.
6. Why are pretreatment and maintenance important in reverse osmosis?
7. Which measurements should trigger an automatic alarm or plant shutdown?

Application Exercise

A small plant treats 500 m^3 of water per day. Raw water has a turbidity of 18 NTU. After sedimentation, turbidity is reduced by 75%, and the filter then removes 92% of the remaining turbidity.

1. Calculate the turbidity after sedimentation and filtration.
2. A target of 0.5 NTU is required. State whether the simplified chain meets it.
3. Calculate the daily volume that must be reprocessed if 3% of production is rejected.
4. Explain why turbidity alone cannot prove microbiological safety.

Engineering Challenge

Design a compact treatment system for a village beside a turbid river. Characterise the hazards, choose the treatment barriers, energy source and monitoring sensors, and define acceptance limits. Describe routine maintenance, reagent and spare-part needs, sludge management, safe storage and the action taken when one measurement exceeds its limit. Compare a simple robust option with a more automated membrane-based option and justify your recommendation.

Speaking Task

Compare a central treatment plant with small local treatment units. Present a four-minute technical recommendation based on water quality, energy, maintenance, resilience, cost and operator skills.

Sources

World Health Organization – Drinking Water: online resource

U.S. Environmental Protection Agency – Drinking Water: online resource

EC15 – Future Transport

How will people and goods move tomorrow?

Learning Objectives

- Compare transport modes using energy, capacity, space and infrastructure criteria.
- Explain the perception–decision–action chain of an autonomous vehicle.
- Design and justify a multimodal transport solution rather than a single vehicle technology.

Engineering News

Battery buses, hydrogen trains, autonomous shuttles and electric aircraft are being tested around the world. The largest environmental gains, however, may come from combining efficient vehicles with public transport, active mobility, higher occupancy and better use of existing infrastructure.

Main Article

Transport engineering must move people and goods safely, reliably and with limited energy and land use. Because journey length, passenger capacity and infrastructure differ, no propulsion technology is ideal for every application.

Battery-electric vehicles are efficient because electric motors convert a large proportion of stored energy into motion and can recover energy during braking. They are well suited to urban buses, cars and delivery vehicles that return regularly to a charging point. Long-distance freight requires larger batteries, faster charging or another energy carrier. Hydrogen fuel cells may reduce onboard storage mass in some heavy applications, but producing, compressing and converting hydrogen introduces additional energy losses.

Rail carries many passengers or large freight loads with low rolling resistance. Electrified trains can use energy directly from the grid and do not need to carry a large energy store. High-speed rail can replace some short-haul flights when stations connect city centres and total door-to-door time is competitive.

Autonomous vehicles add a digital engineering challenge. Cameras provide detailed images, radar measures distance and relative speed, and lidar creates a three-dimensional representation. Sensor fusion combines these signals. Software identifies objects, predicts their motion, plans a safe trajectory and commands steering and braking. Adverse weather, unusual road layouts and human behaviour make validation difficult.

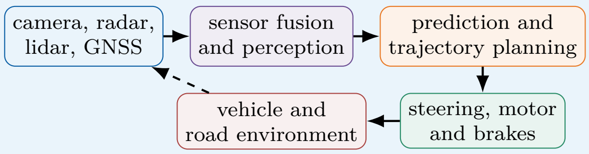
Future mobility is therefore a system-level problem. A journey may combine walking, bicycle, train and an on-demand shuttle. Ticketing, timetables and physical interchanges must work together. Engineers must consider accessibility, affordability, charging networks, manufacture and the carbon intensity of the energy supply. Replacing every conventional car with another type of car would not, by itself, eliminate congestion, parking demand or unequal access to mobility.

Engineering Insight

Vehicle efficiency and transport efficiency are not identical. Occupancy, route, waiting time, vehicle mass and infrastructure use can matter as much as propulsion efficiency. The correct unit of comparison is often energy or emissions per passenger-kilometre or tonne-kilometre.

Engineering Focus

Autonomous driving control chain

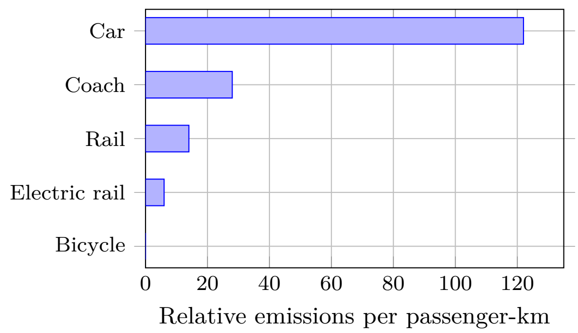


The loop must continue safely when a sensor is degraded. Redundancy, plausibility checks, driver or remote supervision and a minimum-risk manoeuvre are central design features.

Key Figures

Electric motor efficiency	high
Rail rolling resistance	low
Autonomous control loop	continuous
Useful comparison metric	passenger-km
Main urban constraint	limited space

Illustrative operational emissions



Illustrative comparison only. Actual emissions depend strongly on occupancy, vehicle, energy source, route and lifecycle boundaries.

Multimodal journey

home → walking / bicycle → rail → shared shuttle → destination

A successful system minimises waiting, transfer distance, uncertainty and barriers for people with reduced mobility.

Useful Vocabulary

freight	fret
short-haul flight	vol court-courrier
rolling resistance	résistance au roulement
sensor fusion	fusion de capteurs
trajectory planning	planification de trajectoire
minimum-risk manoeuvre	manœuvre de risque minimal
shared mobility	mobilité partagée
interchange	pôle d'échanges
occupancy rate	taux de remplissage
passenger-kilometre	voyageur-kilomètre

Questions

1. Why is no propulsion technology suitable for every journey?
2. Give one advantage of electrified rail over battery vehicles.
3. Describe the autonomous perception–decision–action loop.
4. Why is validation difficult in adverse or unusual conditions?
5. Use the graph to compare the highest- and lowest-emission motorised modes.
6. Why is vehicle occupancy important when comparing transport modes?
7. Which features make a multimodal journey attractive and accessible?

Application Exercise

A class of 30 students travels 18 km to a science museum. Compare five cars carrying six students each with one coach carrying all 30 students, using the illustrative values in the graph.

1. Calculate the total passenger-kilometres for the trip.
2. Estimate the operational emissions for each option in relative units.
3. By what factor does the coach reduce emissions compared with the cars?
4. Recalculate the car option for an average occupancy of three students per car.
5. Give two reasons why a real comparison should also include lifecycle and infrastructure effects.

Engineering Challenge

Propose a low-carbon transport plan between a suburban school and a city centre. Define demand and accessibility requirements; compare at least three modes using capacity, journey time, energy, emissions, land use, reliability and infrastructure; include first- and last-kilometre solutions; and identify one disruption scenario with a fallback plan. Present a weighted decision matrix and a justified recommendation.

Speaking Task

Should city centres restrict private cars? Defend a position in a four-minute technical briefing using transport capacity, accessibility, energy, public space and social impact, and propose one realistic alternative package.

Sources

International Energy Agency – Transport: online resource

European Commission – Mobility and Transport: online resource

EC16 – Cybersecurity

How can engineers protect connected systems?

Learning Objectives

- Define confidentiality, integrity and availability and relate them to physical systems.
- Explain a defence-in-depth architecture for a connected engineering system.
- Build and justify a simple cyber-risk analysis from threats, vulnerabilities and consequences.

Engineering News

Industrial systems, vehicles, hospitals and energy networks increasingly depend on connected software. Cybersecurity is therefore a safety and business-continuity issue as well as an information-technology issue: a digital attack can produce physical consequences.

Main Article

A connected engineering system contains sensors, controllers, software, communication links and actuators. Each interface can become an entry point for accidental failure or malicious action. Cybersecurity aims to preserve three central properties: confidentiality prevents unauthorised disclosure, integrity prevents unauthorised modification, and availability keeps the service usable.

Risk depends on more than the existence of a threat. Engineers identify valuable assets, possible attackers, vulnerabilities, likelihood and consequences. A weak password on a non-critical display is not the same risk as an unprotected remote command to a water-treatment pump. Safety analysis and cyber-risk analysis must therefore be coordinated.

No single protection is sufficient. Defence in depth combines several independent layers. Devices should authenticate users and other devices. Communications can be encrypted, software should be updated, and networks should be segmented so that an office computer cannot directly control a critical machine. Access rights follow the principle of least privilege: each account receives only the permissions required for its task.

Monitoring is also necessary. Logs, alarms and anomaly detection can reveal unexpected traffic or commands. When an incident occurs, operators need a prepared response: isolate affected equipment, maintain a safe physical state, restore trusted software and verify operation. Offline backups help recovery, but they must be tested rather than merely stored.

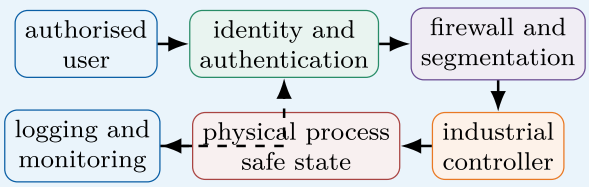
Security creates engineering trade-offs. Encryption and authentication use processing time and energy. Updates may interrupt production, while legacy equipment may not support modern protocols. A secure design therefore begins with requirements and threat modelling rather than with a protective tool added at the end. Human factors matter too: clear procedures and training reduce errors and social-engineering attacks.

Engineering Insight

Safety and cybersecurity can conflict. An emergency door should open quickly for evacuation, yet access control should prevent intrusion. The engineering task is to define priority rules, degraded modes and evidence that both requirements are satisfied.

Engineering Focus

Defence-in-depth architecture

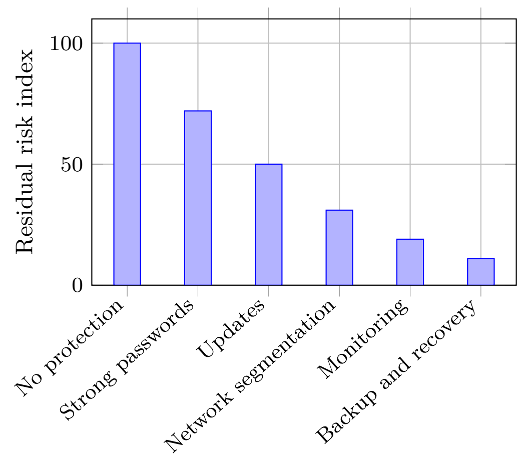


If one layer fails, another should still reduce the probability or consequence of compromise.

Key Figures

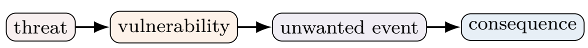
Core security objectives	3
Preferred access principle	least privilege
Critical architecture measure	segmentation
Recovery requirement	tested backup
Essential lifecycle activity	updates

Effect of layered safeguards



Illustrative risk reduction: safeguards are cumulative, but risk never becomes exactly zero and each layer must be maintained and periodically verified.

Risk reasoning



Safeguards may reduce likelihood, limit consequences, improve detection or shorten recovery time.

Useful Vocabulary

confidentiality	confidentialité
integrity	intégrité
availability	disponibilité
threat	menace
vulnerability	vulnérabilité
authentication	authentification
encryption	chiffrement
network segmentation	segmentation du réseau
least privilege	moindre privilège
backup	sauvegarde

Application Exercise

The graph assigns a residual-risk index of 100 to an unprotected system and 11 after all listed safeguards have been introduced.

1. Calculate the percentage reduction in residual risk.
2. Which safeguard produces the largest absolute reduction between two consecutive bars?
3. Explain why the final index is not zero.
4. Rank confidentiality, integrity and availability for a connected greenhouse and justify your order.
5. Propose one verification test for the local fail-safe mode during a network outage.

Questions

1. Define confidentiality, integrity and availability.
2. Why should cyber risk be linked to physical consequences?
3. Give four layers of a defence-in-depth strategy.
4. What is the purpose of network segmentation?
5. Why must backups and incident procedures be tested?
6. Distinguish a preventive safeguard from a recovery safeguard.

Engineering Challenge

Secure a connected greenhouse containing temperature sensors, an irrigation pump and remote access. Identify assets, two threats and two vulnerabilities. Build a small risk matrix, then propose authentication, segmentation, monitoring and a safe local mode. Finish with a verification plan and a justified recommendation to the operator.

Speaking Task

Is a fully connected system always better than an isolated one? Compare functionality, maintenance, security and resilience using one engineering example and one quantified criterion.

Sources

NIST – Cybersecurity Framework: online resource

European Union Agency for Cybersecurity: online resource

EC17 – Internet of Things

How can connected objects remain useful, secure and energy-efficient?

Learning Objectives

- Describe the complete IoT chain from sensing to decision and user service.
- Compare communication technologies using range, data rate, latency and energy criteria.
- Estimate battery life and justify reliability, interoperability, privacy and maintenance choices.

Engineering News

Connected sensors now monitor factories, buildings, farms and transport networks. The engineering challenge is no longer simply to connect a device, but to keep thousands of low-power objects useful, secure and maintainable throughout their entire service life.

Main Article

The Internet of Things (IoT) links physical objects to digital services. A typical object contains a sensor, a microcontroller, a communication interface and a power supply. It measures a physical quantity, processes the signal locally and sends useful information to a gateway or cloud platform. The platform stores data, detects abnormal behaviour and may send a command back to an actuator.

A connected temperature sensor illustrates the complete chain. The sensing element converts temperature into an electrical signal. The microcontroller filters the measurement, adds a timestamp and decides whether transmission is necessary. Sending every value provides detailed data but wastes energy and network capacity. Edge processing can instead transmit only a periodic summary or an alarm when a threshold is exceeded.

Communication technology depends on range, data rate, latency, power and available infrastructure. Bluetooth Low Energy is suited to short-range devices. Wi-Fi offers higher throughput but usually consumes more power. LoRaWAN can transmit small packets over several kilometres, while cellular IoT provides wide-area coverage through an operator network. No protocol is best for every application.

Energy autonomy is often the limiting requirement. Engineers reduce consumption by using sleep modes, low sampling rates and short radio transmissions. A battery-powered sensor may need to operate for several years without maintenance. Energy harvesting from light, vibration or heat can extend battery life, but the available power is small and variable.

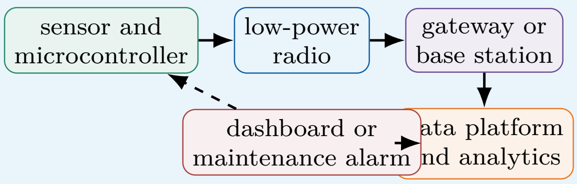
Security and maintainability must be designed from the start. Each object needs a unique identity, authenticated software updates and encrypted communication. A compromised device can reveal private data or become an entry point into a larger system. Interoperability also matters: devices from different manufacturers must use compatible data formats so that the system can evolve without replacing every component.

Engineering Insight

For a battery-powered object, radio transmission may consume far more energy than the measurement itself. The best optimisation may therefore be a change in information strategy: send less often, compress locally or transmit only when the measured state changes significantly.

Engineering Focus

Architecture of a connected monitoring system

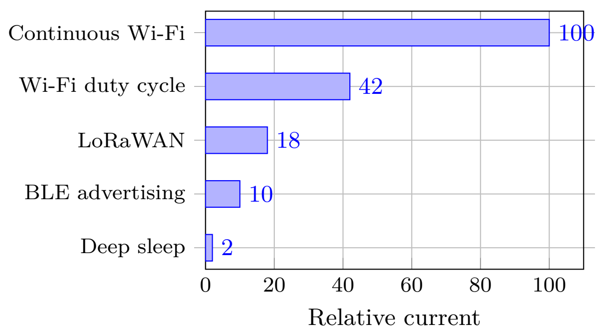


Local processing reduces traffic and allows basic operation even if the cloud connection is lost.

Key Figures

Typical sensor data rate	low
Critical resource	battery energy
Long-range low-power example	LoRaWAN
Short-range example	BLE
Essential lifecycle feature	remote update

Relative communication current



Illustrative values: actual consumption depends strongly on hardware, coverage, transmission duration and retry rate.

Useful Vocabulary

connected object	objet connecté
gateway	passerelle
throughput	débit
sleep mode	mode veille
edge processing	traitement local
packet	paquet de données
energy harvesting	récupération d'énergie
firmware update	mise à jour du micrologiciel
interoperability	interopérabilité
encryption	chiffrement

Application Exercise

A connected sensor is powered by a 2400 mAh battery. During each hour, it remains in deep sleep for 3590 s at a relative current of 2, then transmits for 10 s using LoRaWAN at a relative current of 18. Assume that one relative-current unit corresponds to 0.10 mA.

1. Calculate the average current over one hour.
2. Estimate the ideal battery life in hours and years.
3. Apply a derating factor of 0.70 for temperature, self-discharge and ageing.
4. Estimate the new service life if the device transmits twice per hour.
5. Recommend one design change and justify it quantitatively.

Questions

1. Describe the information chain of an IoT system.
2. Why is edge processing useful for a battery-powered sensor?
3. Compare Wi-Fi, BLE and LoRaWAN for one application.
4. Why are software updates a safety and security requirement?
5. Give two methods for increasing sensor autonomy.
6. Why can poor radio coverage reduce battery life?

Engineering Challenge

Design a connected humidity-monitoring network for a school building. Define requirements for range, autonomy, measurement uncertainty and alarm latency. Select the sensing period, communication method and power source. Add one local fallback function, two privacy measures, a maintenance strategy and a verification plan. Present your final choice in a weighted decision matrix.

Speaking Task

Should every household appliance be connected? Present one useful service, one environmental cost, one cybersecurity risk and one measurable condition that would justify connectivity.

Sources

NIST – Internet of Things: online resource

LoRa Alliance: online resource

EC18 – Drones

How do autonomous aerial systems remain stable and safe?

Learning Objectives

- Explain how thrust, attitude and position are controlled in a multirotor drone.
- Relate sensor characteristics and control-loop bandwidth to flight stability.
- Build a safe mission plan from payload, endurance, weather and fault-response requirements.

Engineering News

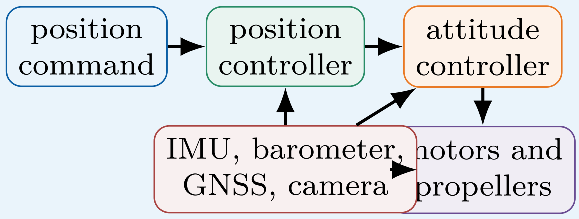
Drones are increasingly used for inspection, mapping, emergency response and precision agriculture. Their usefulness depends on more than flight performance: safe autonomy requires reliable sensing, energy management, communication and clearly defined procedures for fault conditions.

Engineering Insight

A quadrotor normally changes direction by tilting. Part of the total thrust then acts horizontally, so the motors must generate additional thrust to maintain altitude. Aggressive manoeuvres therefore consume energy faster than steady flight even when the travelled distance is unchanged.

Engineering Focus

Nested control loops of a drone



Fast attitude control stabilises the aircraft; slower position control generates the attitude command.

Main Article

A multirotor drone flies by controlling the thrust produced by several propellers. When the total upward thrust equals the weight, the aircraft hovers. Increasing all rotor speeds causes a vertical acceleration. Changing the balance between rotors creates roll, pitch or yaw moments and allows the drone to move horizontally or rotate.

The flight controller closes several feedback loops. Gyroscopes measure angular velocity and accelerometers measure specific acceleration. Their signals are combined to estimate attitude. A barometer provides altitude information, while satellite navigation and cameras can estimate position and horizontal velocity. Sensor fusion is essential because every sensor has limitations: gyroscopes drift, accelerometers are disturbed by vibration and GNSS signals may be inaccurate or unavailable indoors.

A common controller compares the desired attitude with the estimated attitude and adjusts motor speeds many times per second. The inner loop stabilises angular motion, while slower outer loops control velocity and position. If tuning is poor, the aircraft may oscillate or react too slowly to a gust. Sampling frequency, actuator response and filtering must therefore be considered together.

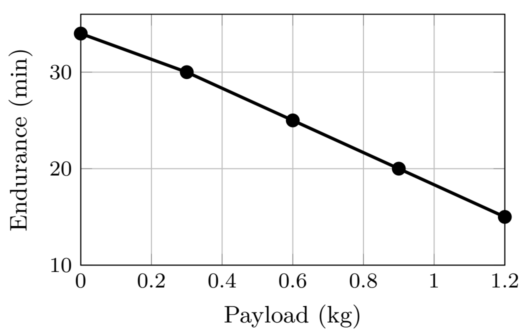
Endurance is strongly limited by battery mass. A larger battery stores more energy but also increases weight and the power required to hover. Payload has the same effect. Engineers therefore optimise the frame, propellers, motors and flight plan together. Wind, temperature and repeated acceleration also reduce real endurance compared with ideal calculations.

Autonomy introduces additional requirements. The drone must detect obstacles, maintain a safe separation from people and execute a predictable action if communication is lost. Depending on the mission, this may be hovering, landing or returning to a predefined point. Geofencing can prevent entry into restricted zones, but it does not replace pilot responsibility, maintenance or risk assessment.

Key Figures

Hover condition	thrust = weight
Fast sensor	gyroscope
Altitude sensor	barometer
Main energy store	battery
Typical fault action	return or land

Payload and endurance



Illustrative trend for one aircraft: endurance decreases non-linearly as payload increases. The graph must not be extrapolated beyond the validated operating range.

Useful Vocabulary

thrust	poussée
hover	vol stationnaire
roll / pitch / yaw	roulis / tangage / lacet
flight controller	contrôleur de vol
gyroscope	gyroscope
sensor fusion	fusion de capteurs
payload	charge utile
gust	rafale
geofencing	géorepérage
return-to-home	retour automatique

Application Exercise

The endurance graph gives 25 min for a payload of 0.6 kg. A safety rule requires the drone to land with 20% of the initial flight time still available. The outward journey to an inspection area takes 6 min and the return journey takes the same time.

1. Calculate the maximum authorised flight time before landing.
2. Determine the time available for inspection over the target area.
3. Recalculate this inspection time for a payload of 0.9 kg.
4. Apply an additional 15% endurance reduction for cold weather and recalculate the mission margin.
5. State whether the mission should be accepted and justify your decision.

Questions

1. How does a quadrotor produce roll and pitch motion?
2. Why is sensor fusion necessary?
3. Distinguish the attitude loop from the position loop.
4. Explain the compromise created by a larger battery.
5. What should happen after a communication failure?
6. Why should a control loop not use an excessively noisy measurement directly?

Engineering Challenge

Define a drone mission to inspect a bridge. Write requirements for payload, image resolution, endurance, wind and separation distance. Select the main sensors, propose a safe path and define the response to low battery, GNSS loss and communication loss. Include an energy budget, a pre-flight checklist and three validation tests before operational approval.

Speaking Task

Should drones be authorised to deliver parcels in cities? Compare efficiency, noise, safety, privacy and infrastructure requirements, then state one quantitative acceptance criterion.

Sources

European Union Aviation Safety Agency – Drones: online resource

NASA – Unmanned Aircraft Systems: online resource

EC19 – Biomimetics

How can engineering learn from living systems?

Learning Objectives

- Distinguish biological inspiration, abstraction and direct imitation.
- Transfer a biological working principle into a manufacturable engineering concept.
- Compare biomimetic and conventional solutions using performance and life-cycle evidence.

Engineering News

Biomimetic research is influencing adhesives, lightweight structures, ventilation systems and robot locomotion. Nature provides useful strategies, but engineering success requires abstraction, modelling and testing rather than simply copying the appearance of an organism.

Main Article

Biomimetics examines biological functions and transfers useful principles to engineered systems. The process usually begins with a technical problem such as reducing drag, gripping a surface or building a lightweight structure. Engineers then search for organisms that solve a comparable functional problem under similar constraints.

The biological solution must be understood at the relevant scale. A bird wing is not only a curved surface; its performance depends on flexible feathers, muscles, sensing and active control. Directly copying its shape may therefore fail. Engineers abstract the underlying principle, create a model and adapt it to available materials and manufacturing methods.

Surface engineering provides many examples. Gecko feet adhere through millions of microscopic hairs that create weak intermolecular forces over a large contact area. Artificial dry adhesives reproduce the hierarchical structure without using glue. Shark skin has rib-like textures that influence the flow close to the surface. Similar riblets can reduce friction drag in selected conditions, although their benefit depends on size, orientation, Reynolds number and cleanliness.

Nature also inspires structures. Bone combines a dense outer layer with a porous interior and achieves a high stiffness-to-mass ratio. Additive manufacturing can produce lattice structures that use the same principle. Honeycomb panels and branching networks similarly offer efficient ways to distribute loads or fluids.

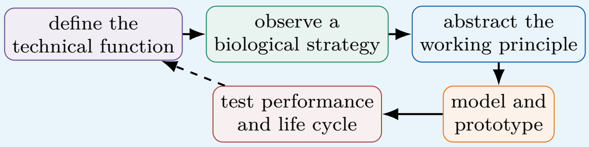
A biomimetic design is not automatically sustainable. Manufacturing a complex microstructure may consume large amounts of energy or use materials that are difficult to recycle. Engineers must compare the complete life cycle, durability and maintenance requirements with a conventional solution. Biomimetics is valuable only when the biological analogy leads to a measurable improvement.

Engineering Insight

The hook-and-loop fastener was inspired by burrs attached to animal fur. Its success came from abstracting a repeatable attachment principle that could be manufactured economically, not from reproducing the complete plant structure.

Engineering Focus

From biological observation to engineering validation

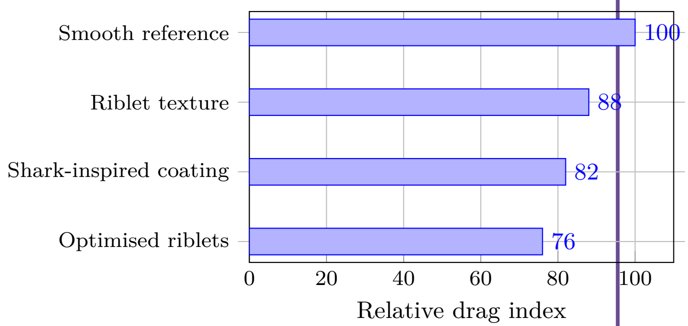


Iteration is essential because biological materials, scales and manufacturing constraints are different.

Key Figures

Primary objective	functional transfer
Common design scale	microstructure
Typical benefit	low mass or drag
Necessary step	abstraction
Final proof	measured validation

Illustrative drag comparison



Illustrative values showing that geometry must be optimised for the operating flow and validated against a smooth reference under identical conditions.

Useful Vocabulary

biomimetics	biomimétisme
living system	système vivant
working principle	principe de fonctionnement
hierarchical structure	structure hiérarchique
riblet	micro-rainure
adhesion	adhérence
porous	poreux
lattice structure	structure en treillis
stiffness-to-mass ratio	rapport rigidité/masse
life cycle	cycle de vie

Application Exercise

A smooth reference surface has a drag index of 100. A biomimetic riblet surface has an index of 82, but its manufacture increases the component mass from 12.0 kg to 12.6 kg and adds 8 impact units.

1. Calculate the percentage reduction in drag index.
2. Calculate the percentage increase in mass.
3. Estimate the net score $S = \Delta D - 2\Delta m$, where reductions are positive percentages.
4. Explain one limitation of this simplified score.
5. Define two controlled experiments and one durability test before approval.

Questions

1. Why is direct imitation often insufficient?
2. Explain how gecko feet inspire dry adhesives.
3. What engineering benefit can be obtained from bone-like structures?
4. Why must operating scale be considered for riblets?
5. How can a biomimetic product have poor environmental performance?
6. What should a fair reference solution include during testing?

Sources

AskNature – Biomimicry examples: online resource
Biomimicry Institute: online resource

Engineering Challenge

Choose a biological strategy that could improve a small wind turbine, robot gripper or protective helmet. Define measurable requirements, abstract the biological principle and propose a manufacturable concept. Compare it with a conventional design using a decision matrix, then specify a prototype, three experiments, acceptance criteria and a life-cycle check.

Speaking Task

Is nature always the best engineer? Discuss optimisation, evolution, materials, manufacturing and the need for measurable evidence. Use one successful and one questionable analogy.

EC20 – Circular Economy

How can products remain useful for longer?

Learning Objectives

- Compare linear and circular product life cycles using delivered service as a reference.
- Explain design for durability, repair, disassembly, refurbishment and remanufacturing.
- Use life-cycle and multicriteria reasoning to avoid shifting environmental impacts.

Engineering News

Manufacturers are developing repairable products, component take-back schemes and remanufacturing lines. Circularity is not achieved by recycling alone: the highest value is often preserved when a product, module or component remains in use with the least possible transformation.

Main Article

The traditional linear model follows a simple sequence: extract resources, manufacture a product, use it and discard it. This model loses materials, embodied energy and technical value at the end of each product life. A circular economy aims to keep products and materials useful for longer through maintenance, reuse, repair, refurbishment, remanufacturing and, finally, recycling. Engineering decisions made during design strongly influence these options. A durable product needs components that resist expected loads and environments. A repairable product needs accessible fasteners, diagnostic information and available spare parts. Modular architecture allows a failed or obsolete function to be replaced without discarding the complete system.

Disassembly must also be considered. Permanent adhesives and mixed materials may simplify production but make separation difficult. Screws, clips and clearly identified polymers can improve recovery, although they may add cost, mass or assembly time. Engineers therefore balance circularity with safety, performance and industrial constraints.

Remanufacturing restores a used product to a defined level of performance. Components are inspected, cleaned, repaired or replaced and then tested. This can preserve more value than material recycling, but it requires traceability and a predictable return flow. Digital product passports can record materials, maintenance history and previous use, helping operators choose the most suitable recovery route.

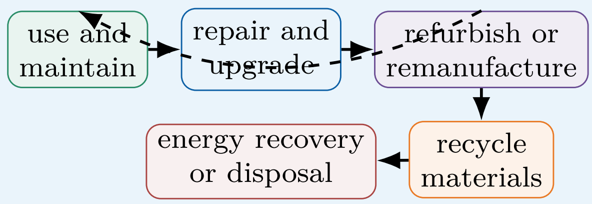
Life-cycle assessment prevents misleading solutions. A heavier product may last longer but require more material and transport energy. A replaceable battery can extend product life, yet the sealing system must still meet safety requirements. Engineers compare resource use, energy, emissions and service life across the complete system boundary. The objective is not merely to close a loop, but to reduce total impact while still providing the required service.

Engineering Insight

Recycling is important, but it often destroys much of the manufacturing value already invested in a component. Repair and reuse usually preserve more value when technical condition, safety and logistics allow them. The shortest loop is not automatically best unless the recovered product still satisfies its requirements.

Engineering Focus

Circular design hierarchy

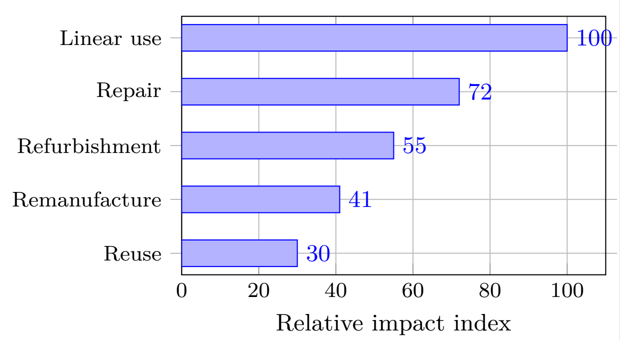


Short loops usually preserve more product value and require less transformation.

Key Figures

Highest-value strategy	continued use
Repair enabler	accessibility
Upgrade enabler	modularity
Recovery enabler	disassembly
Decision tool	life-cycle assessment

Relative life-cycle impact



Illustrative comparison for the same delivered service; real results depend on transport, energy mix, product lifetime, failure rate and replacement parts.

Useful Vocabulary

linear economy	économie linéaire
durability	durabilité
repairability	réparabilité
disassembly	démontage
spare part	pièce détachée
refurbishment	reconditionnement
remanufacturing	remise à neuf industrielle
traceability	traçabilité
product passport	passeport produit
life-cycle assessment	analyse du cycle de vie

Application Exercise

A company compares two strategies for providing 1,000 hours of service. A linear product lasts 500 h and has an impact index of 100 per product. A repairable version lasts 1,000 h and has an initial impact index of 125; one repair adds an impact index of 15.

1. Determine how many linear products are required to provide 1,000 h of service.
2. Calculate the total impact index of the linear strategy.
3. Calculate the total impact index of the repairable strategy, including one repair.
4. Determine the percentage reduction achieved by the repairable strategy.
5. Calculate the break-even repair impact above which the circular strategy is no longer preferable.
6. Give two real-life parameters that should be added to the model.

Questions

1. Why does repair usually preserve more value than recycling?
2. Give three design choices that improve repairability.
3. What is the difference between refurbishment and remanufacturing?
4. Why can permanent adhesive create an end-of-life problem?
5. How does life-cycle assessment prevent impact shifting?
6. Why must alternatives be compared for the same delivered service?

Engineering Challenge

Redesign a cordless drill, laptop or coffee machine for a longer useful life. Define requirements for lifetime, repair time, safety and cost. Propose a modular architecture, reversible joining methods, replaceable components, diagnostics and a collection/remanufacturing process. Compare the original and redesigned products with a weighted decision matrix and identify one rebound or impact-shifting risk.

Speaking Task

Should manufacturers be legally required to provide spare parts and repair information? Discuss cost, safety, innovation, ownership and environmental impact, then propose one measurable obligation.

Sources

European Commission – Circular Economy Action Plan: online resource

Ellen MacArthur Foundation – Circular Economy: online resource

EC21 – Digital Twin

How can a virtual model improve a real system?

Learning Objectives

- Explain the closed information loop between a physical asset and its digital model.
- Distinguish monitoring, diagnosis, prediction and optimisation.
- Evaluate model uncertainty, data quality and validation limits.

Engineering News

Digital twins are used for aircraft engines, production lines, bridges and energy systems. Their value comes from combining measurements with models to estimate hidden states, predict degradation and test decisions before applying them to the real asset.

Main Article

A digital twin is a virtual representation of a physical system that is repeatedly updated with real measurements. Unlike a static CAD model, it evolves during operation. Sensors measure quantities such as temperature, vibration, force, pressure or electrical current. These data are filtered, time-stamped and compared with the behaviour predicted by a model.

The model may be physics-based, data-driven or hybrid. A physics-based model uses conservation laws, material properties and geometry. A data-driven model learns relationships from historical data. A hybrid model combines both approaches: physics constrains the solution while machine learning corrects unknown effects or reduces computation time.

Engineers use twins for several linked functions. Monitoring compares measured and expected behaviour. Diagnosis identifies the probable cause of a residual. Prediction estimates future states, for example the remaining useful life of a bearing. Optimisation compares maintenance dates, control settings or production schedules before they are applied.

A useful twin depends on calibrated sensors, synchronised data and a validated model. A model that is accurate at nominal speed may fail during start-up, overload or extreme weather. Engineers must therefore report uncertainty and define the operating domain in which the twin is trusted.

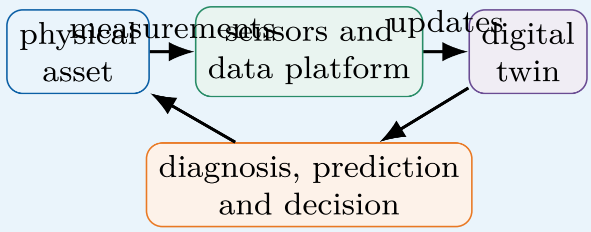
Digital twins do not eliminate physical testing. They help select useful tests, interpret measurements and update decisions throughout the life cycle.

Engineering Insight

A simulation becomes a digital twin only when measurements repeatedly update the model and the model influences decisions about the physical asset.

Engineering Focus

Closed information loop

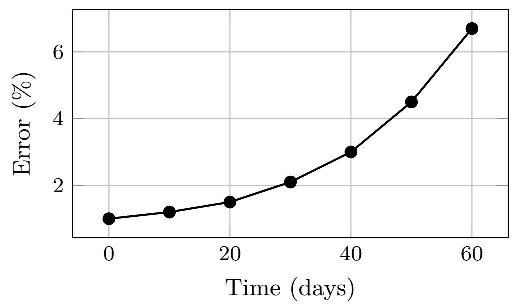


The loop must include validation and human supervision when consequences are safety-critical.

Key Figures

Essential input	calibrated data
Core operation	state update
Typical output	health estimate
Main risk	model drift
Validation method	test comparison

Prediction error over time



A rising residual may indicate wear, sensor drift, a changing environment or an outdated model.

Useful Vocabulary

digital twin	jumeau numérique
asset	équipement réel
state estimation	estimation d'état
residual	résidu / écart
calibration	étalonnage
model drift	dérive du modèle
remaining useful life	durée de vie restante
uncertainty	incertitude
data-driven	fondé sur les données
validation	validation

Questions

1. How is a digital twin different from a CAD model?
2. Compare physics-based and data-driven models.
3. Why must the operating domain of a twin be defined?
4. What can cause a residual to increase?
5. Why does a digital twin not replace physical tests?

Application Exercise

A twin predicts a bearing vibration level of $2.4\,\mathrm{mm\,s^{-1}}$, while the sensor measures $3.0\,\mathrm{mm\,s^{-1}}$.

1. Calculate the absolute residual.
2. Calculate the relative residual as a percentage of the predicted value.
3. The warning threshold is 20%. Decide whether maintenance should be investigated.
4. Give two checks required before concluding that the bearing is damaged.

Engineering Challenge

Design a digital twin for a school ventilation unit. Select four sensors, define the model outputs and one fault indicator, specify the validation tests and explain how uncertainty would be displayed to the operator. Propose one optimisation rule that reduces energy use without lowering indoor-air quality.

Speaking Task

Should a digital twin be allowed to make a safety-critical decision automatically? Discuss reliability, responsibility, uncertainty and human supervision.

Sources

NIST – Digital Twins: online resource

NASA – Digital Twin: online resource

EC22 – Industry 4.0

How can connected factories improve production?

Learning Objectives

- Describe the layered architecture of a connected production system.
- Calculate and interpret availability, performance, quality and OEE.
- Evaluate cybersecurity, traceability and human-factor requirements.

Engineering News

Factories are connecting machines, robots, inspection stations and planning software. The objective is not simply to automate more tasks, but to measure losses, adapt to variation and support faster, better-informed decisions.

Main Article

Industry 4.0 combines connected machines, digital models, automation and data analysis. At machine level, sensors measure cycle time, motor current, vibration, position and product quality. Controllers operate the process in real time, while industrial networks transmit selected information to supervisory systems.

A connected factory is organised in layers. The physical layer contains machines and products. The control layer contains PLCs, drives and robot controllers. A manufacturing execution system coordinates orders, traceability and work instructions. Enterprise software manages resources, suppliers and demand. Information must move between these layers without disturbing deterministic control or production safety.

Condition monitoring can detect a damaged bearing before failure. Vision systems can inspect every part. Production data can reveal that a defect appears only with one tool, one material batch or one temperature range. These applications reduce downtime and waste only when each indicator is connected to a physical cause and a defined action.

Overall equipment effectiveness combines availability, performance and quality:

$$\text{OEE} = A \times P \times Q.$$

A machine with high nominal speed but frequent stops may therefore have poor OEE. Engineers must locate the dominant loss rather than improve the most visible one.

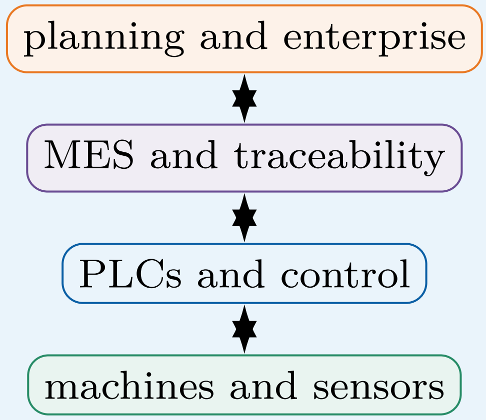
Connectivity also introduces risks. A cyberattack can modify process parameters, legacy equipment may use insecure protocols and operators may receive too many alarms. Successful projects combine network segmentation, clear interfaces, training and human expertise.

Engineering Insight

Collecting more data does not automatically improve production. A useful indicator needs a physical meaning, a decision threshold, an owner and a verification method.

Engineering Focus

Connected production architecture

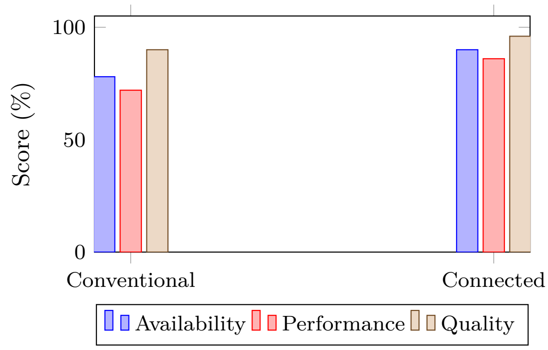


Real-time control should remain local even if an upper information layer becomes unavailable.

Key Figures

Availability loss	downtime
Performance loss	slow cycles
Quality loss	rejected parts
Typical controller	PLC
Critical requirement	secure network

OEE components



Useful Vocabulary

shop floor	atelier de production
PLC	automate programmable
downtime	temps d'arrêt
traceability	traçabilité
work instruction	instruction de travail
condition monitoring	surveillance d'état
quality control	contrôle qualité
throughput	cadence / débit
legacy machine	machine ancienne
cybersecurity	cybersécurité

Questions

1. What are the main layers of a connected factory?
2. How can condition monitoring reduce downtime?
3. What three factors form OEE?
4. Why is collecting all possible data ineffective?
5. Why should machine control remain local?

Application Exercise

During one shift, a line has an availability of 90%, a performance score of 85% and a quality score of 98%.

1. Calculate the OEE.
2. Availability rises to 95%. Calculate the new OEE.
3. Determine the improvement in percentage points and relative percentage.
4. Decide whether improving availability or quality by one additional point produces the larger OEE gain.

Engineering Challenge

A packaging line suffers short, frequent stops. Define a measurement plan, three indicators and their thresholds. Draw the information path from sensor to dashboard, identify one cybersecurity risk and specify how operators will verify that the proposed improvement actually increases OEE.

Speaking Task

Will connected factories create better industrial jobs or reduce human autonomy? Discuss safety, skills, productivity, surveillance and responsibility.

Sources

International Society of Automation: online resource

NIST – Smart Manufacturing: online resource

EC23 – Autonomous Vehicles

How can an autonomous vehicle remain safe?

Learning Objectives

- Describe perception, localisation, prediction, planning and control.
- Explain sensor diversity, redundancy and degraded operating modes.
- Calculate stopping distance and define an operational design domain.

Engineering News

Autonomous systems are being tested in passenger cars, shuttles, mines, ports and warehouses. Progress is fastest in controlled environments, while public roads remain difficult because weather, road geometry, human behaviour and rare events vary widely.

Main Article

An autonomous vehicle must perceive its environment, estimate its own position, predict how other road users may move, plan a safe trajectory and control steering, braking and propulsion. These functions run continuously under strict timing, reliability and safety constraints.

Perception combines cameras, radar, lidar and ultrasonic sensors. Cameras provide colour and detailed shapes but can be affected by darkness, glare or fog. Radar measures distance and relative speed. Lidar produces accurate three-dimensional point clouds but may be sensitive to rain or contamination. Sensor fusion combines complementary information and helps detect inconsistent measurements.

Localisation uses satellite positioning, inertial sensors, wheel odometry and maps. No source is reliable everywhere: GNSS can be blocked in tunnels, while wheel slip creates odometry errors. Redundant estimates allow the system to identify faulty data and enter a safe degraded mode.

The prediction module estimates the future motion of pedestrians, cyclists and vehicles. The planner selects a manoeuvre and generates a trajectory that respects road geometry, traffic rules, vehicle dynamics and comfort. The controller then adjusts steering torque, brake pressure and propulsion.

Total stopping distance includes the distance travelled during sensing, computation and actuator delay, plus the physical braking distance. Because kinetic energy is proportional to v^2, braking distance rises rapidly with speed. Wet roads, worn tyres and downhill gradients reduce available deceleration.

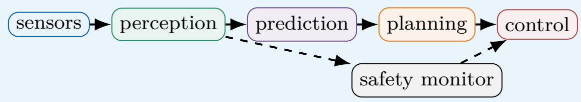
Automation must operate inside a defined operational design domain specifying road type, speed, weather, visibility and map quality. At its boundary, the vehicle must request human control or perform a minimal-risk manoeuvre.

Engineering Insight

Correct object detection is not enough. Safety must be verified across the complete chain from sensing to prediction, decision, control and fallback behaviour.

Engineering Focus

Autonomous driving pipeline

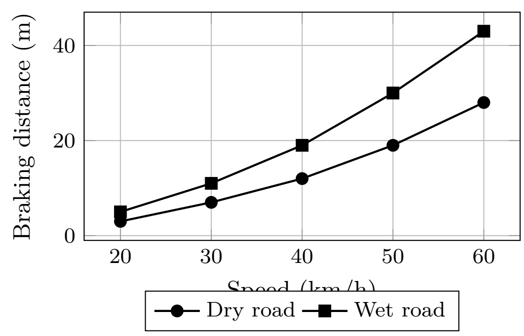


The safety monitor checks plausibility, timing and operational-domain limits independently of the nominal planner.

Key Figures

Camera strength	detailed images
Radar strength	relative speed
Lidar output	3D point cloud
Critical principle	redundancy
Safe fallback	minimal-risk manoeuvre

Illustrative braking distance



Useful Vocabulary

sensor fusion	fusion de capteurs
point cloud	nuage de points
localisation	localisation
trajectory	trajectoire
fallback	solution de repli
operational design domain	domaine opérationnel
wheel slip	glissement des roues
braking distance	distance de freinage
redundancy	redondance
minimal-risk manoeuvre	manœuvre de risque minimal

Questions

1. What functions form the autonomous-driving chain?
2. Why are several sensor technologies combined?
3. How can localisation fail in a tunnel?

4. Why does braking distance rise rapidly with speed?
5. What information belongs in an operational design domain?

Application Exercise

A shuttle travels at $36\,\mathrm{km\,h^{-1}}$. Sensing and computation take $0.35\,\mathrm{s}$ and the brake actuator adds $0.15\,\mathrm{s}$. Assume a deceleration of $5.0\,\mathrm{m\,s^{-2}}$.

1. Convert the speed into $\mathrm{m\,s^{-1}}$.
2. Calculate the distance travelled during the total delay.
3. Using $d_b = v^2/(2a)$, calculate the braking distance.
4. Determine the total stopping distance and a detection range with a 25% safety margin.
5. Repeat the result for a degraded deceleration of $3.0\,\mathrm{m\,s^{-2}}$.

Engineering Challenge

Design an autonomous shuttle for a closed university campus. Define its sensors, maximum speed, operational design domain and minimal-risk manoeuvre. Add a verification matrix covering pedestrians, rain, glare, GNSS loss and one faulty sensor, with a measurable acceptance criterion for each test.

Speaking Task

Who should be responsible after an autonomous-vehicle accident: the passenger, manufacturer, software provider or road operator? Defend a position using technical evidence.

Sources

SAE International – Automated Driving: online resource

NHTSA – Automated Vehicles: online resource

EC24 – Smart Grids

How can electricity networks balance variable resources?

Learning Objectives

- Explain why generation and demand must remain balanced.
- Compare storage, forecasting, demand response and network reconfiguration.
- Calculate an energy balance and define resilience requirements.

Engineering News

Electricity networks are evolving from one-way systems dominated by large power stations into interactive networks containing rooftop solar panels, wind farms, batteries, electric vehicles and flexible loads. Digital control is essential to coordinate these distributed resources.

Main Article

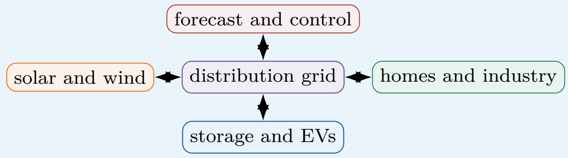
An electricity grid must balance production and consumption at every moment. If generation is lower than demand, system frequency tends to fall; if it is higher, frequency tends to rise. Traditional grids adjusted a limited number of controllable power stations. Smart grids coordinate many smaller, more variable devices.
Smart meters measure energy flows at short intervals. Forecasting software estimates demand and renewable generation. Automated substations can reconfigure part of the network after a fault, while batteries and flexible loads respond rapidly to an imbalance. Demand response shifts selected uses in time: water heating or electric-vehicle charging may be delayed during a peak.
Distributed generation creates bidirectional power flows. A neighbourhood may import electricity in the evening but export solar power at midday. Voltage can rise near photovoltaic installations, and protection systems must distinguish normal reverse flow from a dangerous fault. Engineers therefore combine power electronics, communication systems and network models.
Storage provides different services on different timescales. Batteries smooth short fluctuations, support frequency and reduce peak demand. Pumped-storage hydroelectricity stores much larger amounts for several hours or days. Electric vehicles may provide controlled charging or energy back to the grid, but battery ageing, efficiency and user mobility must be respected.
Digitalisation also increases exposure to cyberattack and communication failure. Local controllers must retain safe behaviour if the central platform is unavailable. Resilience requires physical redundancy, load shedding, black-start capability and plans for extreme weather.

Engineering Insight

Power balance is an instantaneous requirement, whereas energy planning covers minutes, hours, seasons and years. No single storage technology is optimal at every timescale.

Engineering Focus

Bidirectional smart grid

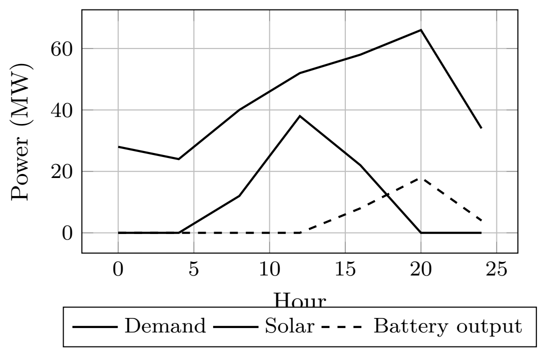


Local protection and fallback rules remain active when communication with the supervisory platform is lost.

Key Figures

Instant requirement	power balance
Fast flexibility	batteries
Long-duration storage	pumped hydro
Flexible demand	smart charging
Critical risk	cyberattack

Daily balancing example



Useful Vocabulary

smart meter	compteur communicant
demand response	effacement / flexibilité
distributed generation	production décentralisée
peak demand	pointe de consommation
frequency control	réglage de fréquence
bidirectional flow	flux bidirectionnel
substation	poste électrique
black start	redémarrage autonome
load shedding	délestage
grid resilience	résilience du réseau

Questions

1. Why must generation equal demand continuously?
2. How does demand response help the network?
3. What problem can midday solar export create?
4. Compare batteries and pumped-storage hydroelectricity.
5. Why should local controllers work without the central platform?

Application Exercise

Between 18:00 and 20:00, a school consumes 140 kW. Its photovoltaic system produces 35 kW, and a battery initially contains 180 kWh of usable energy.

1. Calculate the battery power if no electricity is imported.
2. Determine the energy required during the two-hour period.
3. Decide whether the battery can cover the complete period.
4. Repeat the conclusion with a discharge efficiency of 90% and a reserve of 30 kWh.
5. Propose a demand-response action that would make the requirement achievable.

Engineering Challenge

Plan the energy management of a school with rooftop solar panels, a 200 kWh battery and ten EV chargers. Define priorities for a normal day, a winter peak and a two-hour grid outage. Add a battery state-of-charge rule, a load-shedding sequence and a local fallback mode for communication failure.

Speaking Task

Should electricity suppliers be allowed to delay domestic appliances automatically during peak demand? Discuss consent, comfort, price, privacy and grid stability.

Sources

International Energy Agency – Smart Grids: online resource

ENTSO-E: online resource

EC25 – Marine Renewable Energy

How can oceans supply dependable renewable power?

Learning Objectives

- Compare tidal, wave and offshore energy technologies.
- Explain the mechanical, environmental and maintenance constraints.
- Estimate power, annual energy yield and capacity factor.

Engineering News

Oceans contain major renewable resources. Offshore wind is already deployed at commercial scale, whereas tidal-stream and wave-energy devices are still developing. Marine systems benefit from strong, partly predictable resources but face demanding installation, grid-connection and maintenance conditions.

Main Article

Marine renewable energy includes offshore wind, tidal barrages, tidal-stream turbines and wave-energy converters. Each technology extracts energy from a different physical phenomenon and therefore requires a different structure, control strategy and maintenance plan.

Tidal-stream turbines operate like underwater wind turbines. Because water is much denser than air, even a moderate current can exert large forces. The available kinetic power is

$$P_{\text{avail}} = \frac{1}{2}\rho A v^3,$$

where ρ is water density, A the swept area and v the current speed. The electrical output is lower because rotor, generator and converter efficiencies are limited. Tides are predictable, but suitable sites are limited and the equipment must resist strong cyclic loads.

Wave-energy converters use relative motion between waves and a floating or fixed structure. Some use oscillating water columns; others use hinged bodies or hydraulic power take-off systems. Wave direction, period and height vary continuously, so the controller must adapt while limiting excessive motion. Survival during an extreme storm may be more demanding than normal energy production.

Salt water accelerates corrosion and biological growth. Barnacles and algae modify surface properties and increase drag. Seals, cables and bearings must operate for long periods with limited access. Retrieving a device may require a specialised vessel and a calm weather window, so reliability and maintainability strongly influence the levelised cost of energy.

Engineers must also consider navigation, fishing, underwater noise and marine ecosystems. The best design is therefore not simply the device with the highest peak power, but the system that delivers dependable energy over its complete life cycle.

Engineering Insight

Marine devices are often designed by two operating cases: efficient energy capture in normal conditions and structural survival during rare extreme events.

Engineering Focus

Tidal-stream conversion chain

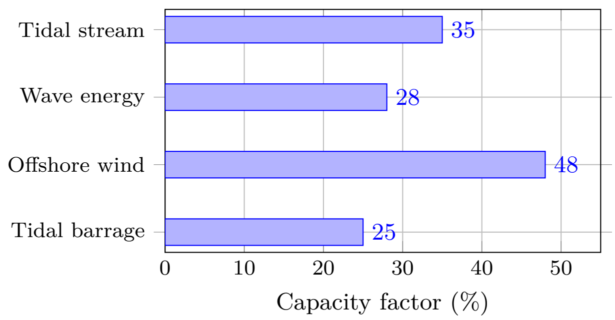
tidal current → rotor → generator → converter → subsea cable and grid

Each conversion stage introduces losses and a possible failure mode that must be inspected or monitored.

Key Figures

Predictable resource	tides
Main material threat	corrosion
Common surface issue	biofouling
Major cost driver	offshore maintenance
Grid connection	subsea cable

Illustrative capacity factors



Illustrative values; real performance depends strongly on the site, availability and technology maturity.

Useful Vocabulary

tidal stream	courant de marée
wave period	période de vague
power take-off	système de conversion d'énergie
cyclic load	charge cyclique
corrosion	corrosion
biofouling	encrassement biologique
subsea cable	câble sous-marin
capacity factor	facteur de charge
mooring	amarrage
storm survival	tenue en tempête

Questions

1. Why can relatively slow seawater flows produce large forces?

2. What makes wave-energy control difficult?
3. Why does maintenance strongly influence marine-energy costs?
4. Give two environmental impacts that must be monitored.
5. Why is peak power insufficient for comparing devices?

Application Exercise

A tidal turbine has a rotor diameter of 18 m in a current of $2.5\,\mathrm{m\,s^{-1}}$. Take $\rho = 1025\,\mathrm{kg\,m^{-3}}$, $C_p = 0.40$ and an electrical efficiency of 90%.

1. Calculate the swept area.
2. Estimate the electrical power produced at this current speed.
3. With a capacity factor of 35%, estimate the annual energy output.
4. The device is unavailable for maintenance for 20 days per year. Recalculate the annual output.
5. Explain why the v^3 relationship makes site measurements critical.

Engineering Challenge

Propose a nominal 1 MW tidal-stream installation. Estimate the rotor area for an assumed current speed, select corrosion and biofouling protection, define condition-monitoring sensors and prepare a maintenance strategy. Add two ecological indicators and one storm-survival verification test.

Speaking Task

Should marine renewable projects be prioritised in protected coastal areas when they reduce carbon emissions? Discuss biodiversity, energy security, local jobs and precaution.

Sources

International Energy Agency – Ocean Power: online resource

European Marine Energy Centre: online resource

EC26 – Biomedical Imaging

Turning physical signals into reliable clinical information

Learning Objectives

- Explain the physical principles and complete signal chain of major imaging techniques.
- Analyse resolution, contrast, noise, artefacts and acquisition-time trade-offs.
- Calculate echo depth and define validation and safety requirements for a clinical protocol.

Engineering News

Biomedical imaging allows clinicians to inspect anatomy and function without surgery. Engineers combine wave physics, detectors, analogue electronics, digital signal processing and inverse algorithms to obtain clinically useful information while controlling acquisition time, cost, uncertainty and patient risk.

Main Article

Biomedical imaging converts interactions inside the body into spatial maps. X-ray radiography and computed tomography measure the attenuation of ionising radiation. Ultrasound transmits mechanical waves and analyses their echoes. Magnetic resonance imaging, or MRI, detects radio-frequency signals produced by hydrogen nuclei in a strong magnetic field. Nuclear medicine detects radiation emitted by a tracer introduced into the body.

Every imaging system contains four linked functions: excitation, interaction, detection and reconstruction. The detector does not usually deliver a ready-to-read image. It records many incomplete and noisy measurements from different positions or times. A reconstruction algorithm estimates the spatial distribution that is most consistent with these measurements and with the physical model of the system.

Image quality cannot be reduced to one number. Spatial resolution determines the smallest visible detail, contrast separates tissues with similar properties, and signal-to-noise ratio indicates whether a weak feature can be distinguished from random fluctuations. Artefacts are structured errors caused by motion, metal, calibration defects, limited viewing angles or an incorrect model. Improving one criterion often degrades another: smaller voxels may increase acquisition time or radiation dose, while faster scans may reduce signal.

In pulse-echo ultrasound, reflector depth is estimated from

$$d = \frac{ct}{2},$$

where c is the speed of sound in tissue and t is the round-trip delay. The factor two accounts for the outward and return paths. Axial resolution depends on pulse duration, whereas lateral resolution depends on beam width and focusing. A protocol must therefore specify both geometry and signal-processing parameters.

Safety constraints depend on the technology. X-ray dose must be kept as low as reasonably achievable. MRI avoids ionising radiation but requires strict control of ferromagnetic objects, radio-frequency heating and acoustic noise. Ultrasound is portable and comparatively inexpensive, yet image quality depends strongly on operator technique and acoustic access.

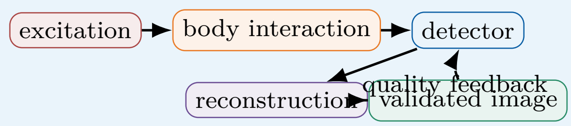
Artificial intelligence can assist segmentation, reconstruction and anomaly detection, but it must be validated on diverse patients, scanners and operating conditions. A plausible-looking image is not automatically a faithful representation of the measurements. Engineers must quantify uncertainty, failure modes and traceability before clinical deployment.

Engineering insight

A reconstructed image is an estimate, not a direct photograph. Its reliability depends on the measurement model, calibration, sampling geometry and regularisation assumptions.

Engineering Focus

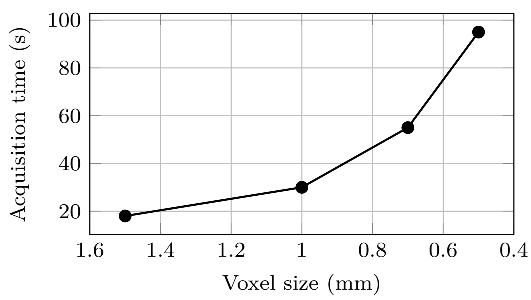
From physical interaction to clinical decision



Key Figures

X-ray / CT	attenuation
Ultrasound	echo delay
MRI	magnetic resonance
Main trade-off	quality–time–risk
Critical requirement	clinical validation

Resolution and acquisition time



Smaller voxels may improve geometric detail, but only if the signal-to-noise ratio remains sufficient.

Application Exercise

Ultrasound depth and uncertainty An echo returns from a tissue boundary after $130\,\mu$s. Take $c = 1540\,\mathrm{m\,s^{-1}}$.

1. Calculate the depth of the boundary.
2. A second echo arrives $26\,\mu$s later. Estimate the distance between the two boundaries.
3. If the assumed sound speed is uncertain by $\pm 20\,\mathrm{m\,s^{-1}}$, estimate the resulting uncertainty on the first depth.
4. Explain why the full travel distance would give a physically incorrect result.

Useful Vocabulary

attenuation	atténuation
echo	écho
voxel	voxel
spatial resolution	résolution spatiale
contrast	contraste
noise	bruit
artefact	artefact
reconstruction	reconstruction
radiation dose	dose de rayonnement
traceability	traçabilité

Questions

1. Why do different imaging technologies reveal different tissue properties?
2. What assumptions are introduced by reconstruction algorithms?
3. Distinguish random noise from a structured artefact.
4. Why is the highest possible spatial resolution not always the best protocol?
5. Give one safety constraint for CT and one for MRI.

Engineering Challenge

Design an imaging protocol for a moving organ. Define the clinical requirement, choose a technology, specify spatial and temporal resolution, identify the dominant artefact, propose one correction method and build a validation matrix containing at least three tests, one acceptance threshold and one degraded operating case.

Speaking Task

Should AI-enhanced medical images be accepted when visibility improves but the original measurements are modified? Discuss accuracy, traceability, responsibility and patient safety.

Sources

World Health Organization – Medical imaging: online resource

International Atomic Energy Agency – Radiation protection: online resource

EC27 – Nanotechnology

Designing materials through scale effects

Learning Objectives

- Explain why surface and quantum effects become dominant at nanometre scales.
- Compare top-down and bottom-up fabrication using accuracy, throughput and defect criteria.
- Quantify surface-area effects and define measurement, containment and lifecycle requirements.

Engineering News

Below about one hundred nanometres, geometry and quantum physics can modify optical, electrical, chemical and mechanical behaviour. Nanotechnology uses these scale effects in electronics, coatings, catalysts, sensors, medicine and energy systems, but performance must be balanced against reproducibility and exposure risk.

Main Article

A nanometre is one billionth of a metre. A nanoscale object may behave differently from the same material in bulk form. For geometrically similar particles, surface area varies with the square of size while volume varies with the cube. The surface-area-to-volume ratio therefore increases as characteristic length decreases. Interfaces, adhesion, oxidation and surface reactions can then dominate the behaviour of the complete object.
For a spherical particle of diameter d,

$$\frac{S}{V} = \frac{6}{d}.$$

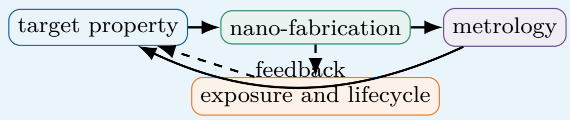
Halving the diameter doubles this ratio. This can increase catalytic activity or dissolution rate, but it may also increase chemical reactivity, agglomeration and biological exposure.
A second scale effect is quantum confinement. When charge carriers are restricted to dimensions comparable with their characteristic wavelengths, allowed energy levels change. Quantum dots can therefore emit different colours according to size, and nanoscale conductors may display transport properties absent from larger components.
Top-down fabrication removes material from a larger structure using lithography, etching or precision machining. It provides accurate patterns and compatibility with semiconductor processes, but cost and defect sensitivity rise as dimensions decrease. Bottom-up fabrication assembles atoms, molecules or particles through synthesis, deposition or self-organisation. It can form very small structures over large areas, although controlling size distribution, orientation and contamination remains difficult.
Characterisation is part of the design loop. Conventional optical microscopy cannot resolve features far below the wavelength of visible light, so electron microscopy, atomic-force microscopy and spectroscopy are used to measure size, morphology, roughness and composition. A nominal particle diameter is insufficient if the batch contains a wide distribution or large agglomerates.
Risk assessment must consider size, shape, surface chemistry, solubility and persistence rather than mass alone. Responsible engineering therefore includes closed handling, local extraction, exposure monitoring, waste control, recycling and end-of-life scenarios. A high-performance nanomaterial is not successful if its manufacture cannot be controlled or verified.

Engineering insight

At the nanoscale, manufacturing tolerance can be of the same order as the designed feature. Metrology and process capability are therefore as important as the nominal geometry.

Engineering Focus

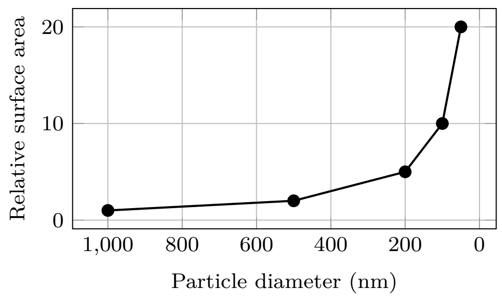
Scale, fabrication and verification



Key Figures

1 nanometre	10^{-9} m
Typical nanoparticle	1–100 nm
Top-down	material removal
Bottom-up	assembly
Critical controls	metrology and exposure

Surface area increases as size decreases



The curve illustrates a geometric scaling law; real performance also depends on accessible surface and agglomeration.

Application Exercise

Particle size and accessible surface A coating is available with spherical particles of diameter 40 nm or 200 nm.

1. Calculate $S/V = 6/d$ for both sizes in m^{-1}.
2. Determine the ratio between the two values.
3. If 30% of the surface of the smaller particles is lost through agglomeration, calculate the effec-

tive gain.
4. Explain one performance benefit and one exposure risk associated with the smaller particles.

Useful Vocabulary

nanoscale	échelle nanométrique
nanoparticle	nanoparticule
thin film	couche mince
lithography	lithographie
etching	gravure
self-assembly	auto-assemblage
quantum dot	boîte quantique
agglomeration	agglomération
coating	revêtement
process capability	capabilité du procédé

Questions

1. Why does the surface-area-to-volume ratio rise as size decreases?
2. Compare one advantage and one limitation of top-down and bottom-up fabrication.
3. Why are electron or atomic-force microscopes often required?
4. Why can agglomeration reduce the expected performance of nanoparticles?
5. Why is particle mass alone insufficient for evaluating health risk?

Engineering Challenge

Specify a nanostructured anti-corrosion coating for an aluminium component. Define the target property, particle size range, fabrication route, two measurable acceptance criteria, one sampling plan, two exposure controls and an end-of-life strategy. Include a short risk–benefit table and justify whether the nanoscale solution is necessary.

Speaking Task

Do the potential benefits of nanoparticles justify deployment before all long-term environmental effects are known? Discuss evidence, precaution, regulation and lifecycle responsibility.

Sources

European Commission – Nanomaterials: online resource

National Nanotechnology Initiative: online resource

EC28 – Quantum Technologies

Controlling fragile states for computing, sensing and communication

Learning Objectives

- Distinguish classical bits from qubits and explain superposition, interference and measurement.
- Analyse coherence time, gate duration, error rate and hardware overhead.
- Compare realistic applications in quantum computing, sensing and secure communication.

Engineering News

Quantum technologies are progressing from laboratory demonstrations towards specialised sensors, secure links and prototype processors. Their engineering value comes from preparing, controlling and measuring individual quantum states with sufficient fidelity, stability and repeatability.

Main Article

A classical bit takes the value 0 or 1. A qubit can be prepared in a state $\alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$. Measurement returns 0 or 1 with probabilities determined by these amplitudes. The amplitudes themselves are not directly observable in a single measurement, so a quantum experiment must be repeated many times to estimate probabilities.

Quantum computation does not simply test every answer at once. Gates modify amplitudes, while interference suppresses some outcomes and reinforces others. A useful algorithm must therefore organise interference so that informative results become more probable before the final measurement. Quantum speed-up is task-dependent and does not make every computation faster.

Several qubits may become entangled, meaning that their joint state cannot be represented as independent classical variables. Entanglement is a resource for algorithms, communication and metrology, but it also increases the complexity of calibration and verification.

The main engineering limitation is decoherence. Unwanted interactions with the environment disturb phase and energy states. Superconducting processors may require millikelvin temperatures and filtered microwave control. Trapped-ion and neutral-atom systems require ultra-high vacuum, stable lasers and precise electromagnetic fields. Photonic systems avoid some cooling constraints but need low-loss sources, circuits and detectors.

If coherence decays exponentially, a simple model is

$$C(t) = C_0 \exp(-t/T_2),$$

where T_2 is the coherence time. This model gives a first estimate of the available operation window, but a real processor is also limited by gate errors, readout errors, cross-talk, leakage and calibration drift.

Quantum error correction encodes one logical qubit in many physical qubits. Syndrome measurements reveal selected errors without directly measuring the logical information. Fault-tolerant operation requires physical error rates below a demanding threshold and a large hardware overhead. Qubit count alone is therefore a poor performance metric; fidelity, connectivity, cycle time and logical error rate are equally important.

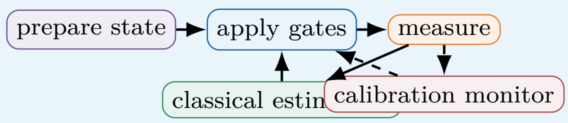
Quantum sensing may reach practical use earlier in several fields. Atomic clocks, magnetometers and gravimeters measure tiny frequency, magnetic-field or acceleration changes. Longer interrogation can improve sensitivity, but also increases exposure to decoherence and environmental drift. Engineering optimisation must include duty cycle, calibration and conventional signal processing.

Engineering insight

A large number of physical qubits does not guarantee useful computation. The relevant question is how many reliable logical operations can be completed before accumulated error makes the result unusable.

Engineering Focus

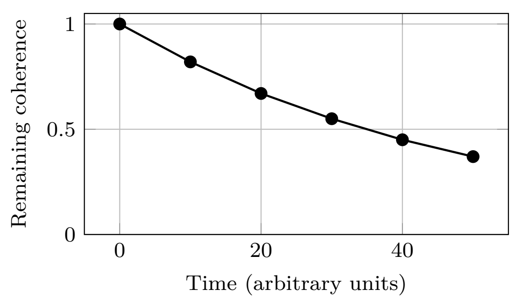
Hybrid quantum–classical control loop



Key Figures

Information unit	qubit
State lifetime	coherence time
Operation quality	gate fidelity
Protected information	logical qubit
Near-term strength	sensing

Illustrative coherence decay



The exponential model is useful for first estimates but does not represent every noise mechanism.

Application Exercise

Coherence budget A qubit has $C_0 = 1$ and $T_2 = 80\,\mu$s. Assume $C(t) = \exp(-t/T_2)$.

1. Calculate the remaining coherence after $40\,\mu$s

and 120 μs.

2. Determine the maximum time for which $C(t) \geq 0.70$.
3. A gate lasts 50 ns. Estimate the theoretical number of sequential gates in this interval.
4. If each gate has fidelity 99.9%, estimate the probability that 500 gates are all error-free.
5. Explain why both answers are optimistic upper bounds.

Useful Vocabulary

qubit	qubit
superposition	superposition
entanglement	intrication
interference	interférence
measurement	mesure
decoherence	décohérence
quantum gate	porte quantique
error correction	correction d'erreurs
gate fidelity	fidélité d'une porte
fault-tolerant	tolérant aux fautes

Questions

1. How does a qubit differ from a classical bit?
2. Why must a quantum experiment be repeated to estimate probabilities?
3. What physical mechanisms can cause decoherence?
4. Why are many physical qubits needed for one logical qubit?
5. Why is qubit count alone an insufficient benchmark?

Engineering Challenge

Compare three candidate technologies for a quantum sensor: superconducting circuits, trapped atoms and photonics. Define five weighted criteria, including sensitivity, operating environment, calibration, maturity and cost. Build a decision matrix, identify one dominant risk for the selected technology and propose a validation experiment with an acceptance threshold and a conventional reference sensor.

Speaking Task

Should governments prioritise quantum computing, quantum sensing or quantum-secure communication? Defend one investment strategy using maturity, expected social benefit, infrastructure cost and technological sovereignty.

Sources

NIST – Quantum information science: online resource

European Quantum Flagship: online resource

EC29 – Sustainable Buildings

Balancing comfort, energy and lifecycle carbon

Learning Objectives

- Distinguish operational energy, embodied carbon and whole-life performance.
- Apply heat-loss and heat-pump calculations to a simple building envelope.
- Define commissioning, monitoring and acceptance criteria for a low-energy building.

Engineering News

Buildings create environmental impacts during material production, construction, operation, maintenance and end of life. Sustainable design combines passive architecture, efficient systems, low-carbon materials, renewable energy and measured verification rather than relying only on simulation labels.

Main Article

A sustainable building must provide comfort, safety, indoor-air quality and useful space while reducing impacts over its complete lifecycle. Operational energy is used for heating, cooling, ventilation, lighting and equipment. Embodied carbon is associated with raw materials, manufacturing, transport, construction, replacement and demolition. As envelopes improve and electricity becomes less carbon-intensive, embodied impacts can represent a larger share of total emissions.

The first design principle is to reduce demand. Transmission through one envelope element can be estimated from

$$\dot{Q} = UA\Delta T,$$

where U is thermal transmittance, A is area and ΔT is the indoor–outdoor temperature difference. Lower U-values, fewer thermal bridges and improved airtightness reduce winter losses. Orientation, glazing ratio, external shading and thermal mass influence solar gains and summer overheating.

Airtightness must be combined with controlled ventilation. Without sufficient air renewal, humidity and pollutant concentrations can rise. Demand-controlled ventilation may use occupancy and carbon-dioxide measurements, but CO_2 is mainly an indicator of occupancy-related ventilation and not a complete measure of indoor-air quality. Particles, volatile organic compounds, moisture and outdoor pollution may require additional sensing or filtration.

Heating and cooling equipment must be selected after the demand has been reduced. A heat pump delivers heat using electrical work, with

$$\mathrm{COP} = \frac{Q_{\mathrm{heat}}}{W_{\mathrm{electric}}}.$$

COP decreases as the temperature lift increases. Low-temperature emitters, good source conditions and a well-insulated envelope therefore improve system performance. Seasonal performance is more relevant than a single laboratory COP.

Material decisions require lifecycle assessment. Concrete offers durability and thermal mass but cement production is carbon intensive. Timber can reduce structural mass and store biogenic carbon, yet sourcing, fire, moisture and end-of-life scenarios must be assessed. Reusing structural components can avoid both demolition waste and new manufacturing.

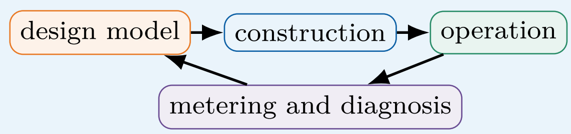
A digital model remains a prediction. The performance gap between calculated and measured consumption can result from construction defects, incorrect controls, unexpected occupancy, weather or user behaviour. Commissioning, sub-metering, trend analysis and corrective action are therefore part of the engineering process. A building is successful only if it continues to satisfy requirements in operation.

Engineering insight

The lowest design consumption is not useful if controls are poorly commissioned. Requirements must therefore include measurable operational indicators, not only simulated annual energy.

Engineering Focus

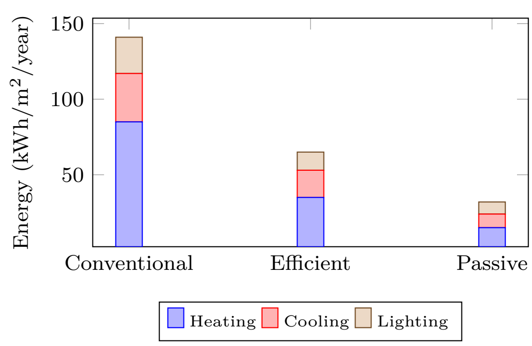
Building performance loop



Key Figures

First priority	reduce demand
Envelope equation	$\dot{Q} = UA\Delta T$
Heat-pump metric	seasonal COP
Air-quality proxy	CO_2
Operational proof	commissioning and metering

Illustrative annual energy demand



Application Exercise

Classroom heat-loss budget A classroom has 120 m^2 of wall with $U = 0.25\ \mathrm{W\,m^{-2}\,K^{-1}}$ and 30 m^2 of windows with $U = 1.4\ \mathrm{W\,m^{-2}\,K^{-1}}$. The indoor–outdoor temperature difference is 20 K.

1. Calculate the transmission heat loss through walls and windows.
2. Add a ventilation loss of 0.65 kW and a thermal-bridge allowance equal to 12% of transmission loss.
3. A heat pump has COP = 3.5. Estimate its electrical input power.
4. Propose one measured indicator that would reveal a commissioning problem after occupancy.

Useful Vocabulary

embodied carbon	carbone incorporé
operational energy	énergie d'exploitation
airtightness	étanchéité à l'air
thermal bridge	pont thermique
shading	protection solaire
heat pump	pompe à chaleur
commissioning	mise au point
sub-metering	sous-comptage
performance gap	écart de performance
lifecycle assessment	analyse du cycle de vie

Questions

1. What is the difference between operational and embodied carbon?
2. Why should demand reduction precede equipment selection?
3. How can external shading improve summer comfort?
4. Why can actual consumption differ from simulation?
5. Why is CO_2 useful but insufficient as a complete air-quality indicator?

Engineering Challenge

Design a low-energy classroom for a temperate climate. Set envelope, shading, ventilation and heating requirements; calculate a peak heat-loss estimate; compare two structural options using embodied carbon; define five sensors and three one-year performance indicators; and specify an acceptance test for comfort, air quality and energy use.

Speaking Task

Should building regulations limit embodied carbon as strictly as operational energy? Discuss measurement uncertainty, affordability, renovation and lifecycle responsibility.

Sources

International Energy Agency – Buildings: online resource

UN Environment Programme – Buildings and construction: online resource

EC30 – High-Speed Transportation

Optimising speed, energy, capacity and safety

Learning Objectives

- Explain how aerodynamic drag, propulsion power and braking distance scale with speed.
- Compare conventional rail, magnetic levitation and low-pressure concepts at system level.
- Estimate stopping distance and define energy, infrastructure and safety requirements for a corridor.

Engineering News

High-speed rail, magnetic-levitation systems and low-pressure-tube concepts seek to reduce journey times between cities. At high speed, aerodynamic resistance, guideway accuracy, signalling, traction power and emergency procedures become dominant constraints. The vehicle and infrastructure must therefore be designed as one system.

Main Article

High-speed transportation is not simply a conventional vehicle with a larger motor. Track geometry, propulsion, power supply, signalling, stations, maintenance and evacuation must be designed together. At $300\,\mathrm{km\,h^{-1}}$, a train travels at $83.3\,\mathrm{m\,s^{-1}}$, so a five-second delay adds more than four hundred metres to stopping distance before effective braking even begins.

Aerodynamic drag can be approximated by

$$F_D = \frac{1}{2}\rho C_D A v^2,$$

where ρ is air density, C_D the drag coefficient, A frontal area and v speed. The associated power is

$$P_D = F_D v = \frac{1}{2}\rho C_D A v^3.$$

Thus, doubling speed multiplies aerodynamic power by approximately eight if geometry and air density remain unchanged. Streamlined noses, sealed inter-car gaps and protected underbodies reduce drag and pressure fluctuations. Tunnel entry is especially demanding because the train compresses air in a confined space.

Steel-wheel high-speed rail offers low rolling resistance, mature switching technology and partial compatibility with existing networks. Magnetic levitation removes wheel–rail contact at speed and can reduce mechanical wear, but requires dedicated guideways, accurate air-gap control and specialised power electronics. Low-pressure tubes could reduce aerodynamic loss further, although maintaining a sealed structure, controlling thermal expansion and providing safe evacuation are major unresolved system constraints.

For constant deceleration a, ideal braking distance is

$$d_b = \frac{v^2}{2a}.$$

Real stopping distance also contains sensing, communication, decision and brake-build-up delays. High-speed lines therefore use continuous cab signalling and automatic train protection. Safe separation must include uncertainty in adhesion, gradient, train mass, actuator response and localisation.

Regenerative braking can return electrical energy when the supply network, another train or a storage system can absorb it. Recovery is limited by conversion efficiency, timetable coincidence, grid receptivity and auxiliary loads. An energy balance must also include rolling resistance, gradients, acceleration, HVAC and station operation.

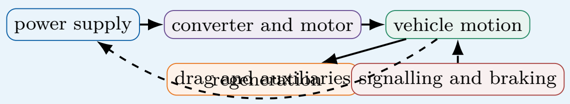
Maximum speed alone is a poor indicator of usefulness. Door-to-door time includes access, waiting, dwell and transfer times. Capacity, reliability, noise, construction carbon and passenger demand can dominate the lifecycle assessment. A high-speed corridor is justified only when system benefits exceed the infrastructure, energy and maintenance costs over decades of operation.

Engineering insight

A higher maximum speed may save little time when stations are close together. Acceleration, braking and dwell time must be included before selecting the design speed.

Engineering Focus

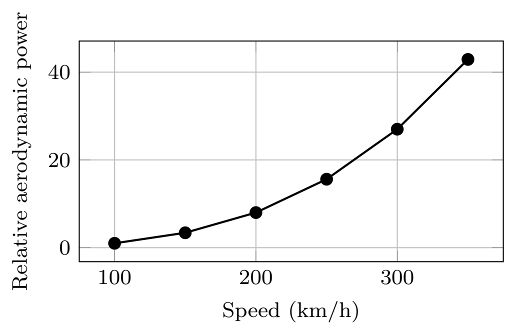
System-level energy and safety chain



Key Figures

Drag force	$\propto v^2$
Aerodynamic power	$\propto v^3$
Ideal braking distance	$v^2/(2a)$
Safety architecture	continuous protection
Useful metric	passenger-kilometre

Aerodynamic power versus speed



The cubic trend explains why modest increases in cruise speed can require large increases in traction energy.

Application Exercise

Stopping-distance and power budget A train has $C_D A = 8.0\,\mathrm{m}^2$ and runs in air of density $1.2\,\mathrm{kg\,m^{-3}}$ at $300\,\mathrm{km\,h^{-1}}$.

1. Convert the speed to $\mathrm{m\,s^{-1}}$ and estimate aerodynamic drag and power.
2. With emergency deceleration $0.9\,\mathrm{m\,s^{-2}}$, calculate the ideal braking distance.
3. Add a five-second response delay and determine the total stopping distance.
4. Apply a 15% safety margin and propose a minimum protected distance.
5. Estimate the aerodynamic power at $330\,\mathrm{km\,h^{-1}}$ using the cubic scaling law and discuss the time–energy trade-off.

Useful Vocabulary

aerodynamic drag	traînée aérodynamique
rolling resistance	résistance au roulement
guideway	voie de guidage
magnetic levitation	sustentation magnétique
pressure wave	onde de pression
braking distance	distance de freinage
regenerative braking	freinage régénératif
train protection	protection automatique des trains
passenger-kilometre	passager-kilomètre
dedicated infrastructure	infrastructure dédiée

Questions

1. Why does aerodynamic power rise more quickly than drag force?
2. Compare the main advantages and limitations of magnetic levitation.
3. Why are continuous signalling and automatic protection necessary?
4. Under what conditions can regenerative braking recover useful energy?
5. Why can maximum speed be a poor measure of door-to-door performance?

Engineering Challenge

Compare a $300\,\mathrm{km\,h^{-1}}$ railway and a magnetic-levitation line for a 600 km corridor. Build a requirement table covering journey time, capacity, energy, protected stopping distance, guideway tolerance, evacuation and availability. Estimate non-stop travel time and aerodynamic power, identify three degraded modes, and propose a lifecycle decision matrix including infrastructure carbon and passenger demand.

Speaking Task

Should public investment favour maximum speed or a denser network with lower speeds and more frequent services? Discuss accessibility, capacity, energy, regional equality and lifecycle cost.

Sources

International Union of Railways – High-speed rail: online resource

International Energy Agency – Rail: online resource